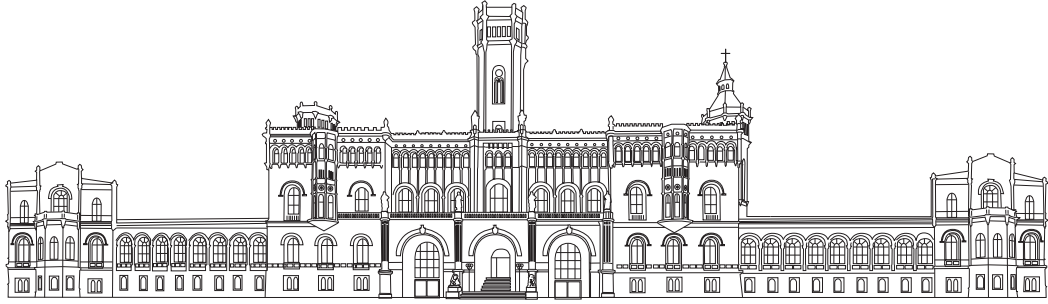




LEIBNIZ UNIVERSITÄT HANNOVER

FAKULTÄT FÜR ELEKTROTECHNIK UND INFORMATIK
INSTITUT FÜR KOMMUNIKATIONSTECHNIK

Towards The Optimal Utility Of Link Level Quantum Communication Systems

Bachelorarbeit

eingereicht von

NICK JONATHAN HERMANN

am 02.04.2026

Erstprüfer : Prof. Dr. Amr Rizk
Zweitprüfer : Prof. Dr.-Ing. Markus Fidler
Betreuer : M.Sc. Karim Elsayed

EHRENWÖRTLICHE ERKLÄRUNG

Hiermit versichere ich, die vorliegende Bachelorarbeit ohne Hilfe Dritter und nur mit den angegebenen Quellen und Hilfsmitteln angefertigt zu haben. Alle Stellen, die wörtlich oder inhaltlich aus den Quellen entnommen wurden, sind als solche kenntlich gemacht worden. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Hannover, 02.04.2026

Nick Jonathan Hermann

ABSTRACT

Quantum communication networks rely on entangled qubit pairs as a key resource for quantum teleportation. Managing these resources is challenging due to stochastic generation, decoherence, and probabilistic purification. This leads to a sequential decision-making problem balancing teleportation fidelity and availability. In this work, we model entanglement management as a Markov Decision Process and apply reinforcement learning to learn adaptive policies using multiple purification protocols. Our results show that reinforcement learning outperforms heuristic strategies in regimes with sufficiently high coherence times. However, we find that increasing the number of available purification protocols does not improve performance in environments with fixed error types, as policies converge to a minimal set of effective actions. Additionally, a single model trained on random errors generalizes well but does not always match specialized models.

ZUSAMMENFASSUNG

Quantennetzwerke nutzen verschränkte Qubit-Paare als zentrale Ressource für die Quantenteleportation. Die Verwaltung dieser Ressourcen ist aufgrund stochastischer Erzeugungsprozesse, Dekohärenz und probabilistischer Purification-Protokolle eine Herausforderung. Dadurch entsteht ein sequenzielles Entscheidungsproblem, bei dem ein Gleichgewicht zwischen Teleportationsfidelity und Verfügbarkeit gefunden werden muss. In dieser Arbeit modellieren wir das Entanglement-Management als Markov-Entscheidungsprozess und verwenden Reinforcement Learning, um adaptive Strategien unter Nutzung mehrerer Purification-Protokolle zu erlernen. Unsere Ergebnisse zeigen, dass Reinforcement Learning heuristische Strategien in Regimen mit ausreichend hohen Kohärenzzeiten übertrifft. Allerdings stellen wir fest, dass eine Erhöhung der Anzahl verfügbarer Purification-Protokolle in Umgebungen mit festen Fehlertypen keine Leistungsverbesserung bringt, da die gelernten Strategien auf eine minimale Menge effektiver Aktionen konvergieren. Darüber hinaus generalisiert ein einzelnes Modell, das auf zufälligen Fehlerverteilungen trainiert wurde, gut, erreicht jedoch nicht in allen Fällen die Leistung spezialisierter Modelle.

CONTENTS

1	Introduction	1
1.1	Motivation	1
1.2	Problem Statement And Contribution	2
2	Background	5
2.1	Quantum Communication	5
2.1.1	Qubits	5
2.1.2	No Cloning Theorem And Measurement Postulate	5
2.1.3	Multiple Qubits	6
2.1.4	Quantum Teleportation	8
2.1.5	Performance Measurement	10
2.1.6	Purification	10
2.2	Deep Reinforcement Learning	11
2.2.1	Policy Gradient Methods	13
2.2.2	Neural Networks	13
2.2.3	Proximal Policy Optimization	13
3	Related Work	15
3.1	Entanglement Purification And Distillation	15
3.2	Learning-Based Control	15
4	Optimal Policy Using Reinforcement Learning	17
4.1	Model	17
4.2	Problem Definition And Approach	20
4.2.1	Modeling As Markov Decision Process	20
4.2.2	Using Proximal Policy Optimization	22
4.3	Implementation	23
4.3.1	Environment	23
4.3.2	Agent	24
4.3.3	Agent Implementation and Hyperparameters	24
4.4	Learning Process	24
5	Evaluation	27
5.1	Baseline	27
5.1.1	Entanglements With Single Error Type	27
5.1.2	Entanglements With Multiple Error Types	28
5.1.3	Influence Of Decoherence On Action Performance	30
5.2	Optimal Policy	30
5.2.1	Evaluating For Different Coherence Times And Error Types	32
5.2.2	4-GPS For Different Error Types	37
5.2.3	Evaluating For Different Wait Time Sensitivities	41
6	Conclusion	45
	Bibliography	47

LIST OF FIGURES

Figure 2.1	Quantum gates and their properties	7
Figure 2.2	Agent-Environment-Interface	12
Figure 4.1	System Model	17
Figure 4.2	Timeline of asynchronous system events. Application qubits and entanglements arrive independently. The system can perform different operations given an entanglement in short-term and long-term memory. In this example, purification was applied.	19
Figure 4.3	The agent interacting with the MPD (simplified, does not show all things that can happen). Points out that multiple events can occur in one transition from one step to another, without the agent seeing the event-loop from the outside. The agent only sees the environment through the state and reward, whenever the environment needs a decision from the agent, not whenever an event happens. Also shows that reward is only other than 0 if a qubit is teleported and then, the episode ends instantly.	23
Figure 5.1	Jump function, success probability and avg. teleportation fidelity for entanglements E with $F(E) = 0.7$ and $\lambda_{active} = 0.3$	29
Figure 5.2	Comparison of the performance of $\lambda_{matched} = 0.2, 0.1$ or 0.0	29
Figure 5.3	(a) shows 5-GPS vs 2-GPS (PMD) performance. (b) shows 4-GPS vs 2-GPS (PROT_2) performance. Both include baseline strategies.	32
Figure 5.4	Action probabilities for 5-GPS for coherence times 0.01 and 0.03).	33
Figure 5.5	F_l threshold upon which KEEP BEST is used for 5-GPS for different coherence times t_c and time since last request t_r values	34
Figure 5.6	Demonstration of the "Push and Hold" strategy applied by the 5-GPS model in a simulation	35
Figure 5.7	Action probability values of the models 5-GPS and 2-GPS (PMD) for $t_c = 0.05$ and different values of F_l and t_r	36
Figure 5.8	Action probability values for $E(0.3, 0.0, 0.0)$ and 2-GPS (PROT_2) at $t_c = 0.05$ across different values of F_l and t_r	37
Figure 5.9	Comparison of 4-GPS performance for alternative error models. The baseline strategies are fully depicted and remain consistent across all evaluated scenarios.	38
Figure 5.10	Action probabilities for the 4-GPS models, depending on F_l and time since last request for entanglements with different errors and $t_c = 0.05$	39
Figure 5.11	Comparison of 4-GPS performance for alternative error models.	40
Figure 5.12	Comparison of 4-GPS performance for alternative error models. The baseline strategies are fully depicted and remain consistent across all evaluated scenarios.	41

Figure 5.13	Action probabilities for 4-GPS (Random) for $t_c = 0.05$. The lines demonstrate how the agent shifts its policy from purification to replacement as the request ages (t_{req} increases), regardless of the error type (λ).	42
Figure 5.14	Comparison of teleportation fidelity across different waiting time sensitivities κ . The 5-GPS and 2-GPS models maintain superior stability compared to simpler protocols.	42

LIST OF TABLES

Table 2.1	Operations Bob needs to apply	9
Table 4.1	Overview of the pumping protocols with their jump functions and success probabilities.	19
Table 5.1	Simplified jump functions and success probabilities for PROT_1-3	28
Table 5.2	PPO Hyperparameter	31
Table 5.3	Model Configurations	31

INTRODUCTION

1.1 MOTIVATION

Quantum communication networks enable the transmission of quantum information between distant nodes and form the foundation for future technologies such as distributed quantum computing [10], secure communication [20], and quantum key distribution [15]. A fundamental mechanism for information transfer in such networks is quantum teleportation, which allows the transmission of qubits without physically sending the particle that carries the quantum state [9].

Quantum teleportation relies on pre-shared entangled qubit pairs between communicating nodes. These entangled pairs serve as a critical resource for teleporting qubits sent by the applications using the network, hereafter referred to as application qubits. These entanglements must be generated and maintained throughout the operation of the network. However, entanglement generation is inherently probabilistic and may fail, due to an entangled qubit being absorbed when traveling to a distant node. If this happens, a new attempt is needed [13]. Furthermore, both entangled states and application qubits are subject to decoherence, which leads to a degradation of their quality over time [9].

The quality of quantum teleportation is commonly measured using fidelity, which quantifies how accurately the transmitted state matches the original one. This fidelity depends on both the quality of the entangled resource and the state of the application qubit at the time of teleportation. In addition, application qubits may need to wait for entanglement to become available, during which their quality further degrades due to decoherence.

To counteract the degradation of entangled states, purification protocols can be applied. These protocols combine multiple low-quality entangled pairs to produce fewer pairs with higher fidelity [5, 16]. More advanced purification schemes and their performance characteristics have been studied in subsequent work, including generalized protocol families and analytical performance models [13, 23]. However, purification is a probabilistic process and may fail, leading to the loss of all involved resources. As a result, there exists a fundamental trade-off between fidelity and availability in quantum communication systems [13, 19].

Alternatively, entanglement can be replaced by generating new pairs, which reduces the risk of complete resource loss but may not meet the fidelity requirements of quantum applications. These challenges highlight the need for intelligent and adaptive strategies that can dynamically balance fidelity and availability in quantum communication systems.

Quantum communication networks enable the transmission of quantum information between distant nodes and form the foundation for future technologies such as distributed quantum computing [10], secure communication [20], and quantum key distribution [15]. A fundamental mechanism for information transfer in such networks is quantum teleportation, which allows the transmission of qubits without physically sending the particle that carries the quantum state [9].

Quantum teleportation relies on pre-shared entangled qubit pairs between communicating nodes. These entangled pairs serve as a critical resource for teleporting qubits sent by the applications using the network, hereafter referred to as application qubits. These entanglements must be generated and maintained throughout the operation of the network. However, entanglement generation is inherently probabilistic and may fail, due to an entangled qubit being absorbed when traveling to a distant node. If this happens, a new attempt is needed [13]. Furthermore, both entangled states and application qubits are subject to decoherence, which leads to a degradation of their quality over time [9].

The quality of quantum teleportation is commonly measured using fidelity, which quantifies how accurately the transmitted state matches the original one. This fidelity depends on both the quality of the entangled resource and the state of the application qubit at the time of teleportation. In addition, application qubits may need to wait for entanglement to become available, during which their quality further degrades due to decoherence.

To counteract the degradation of entangled states, purification protocols can be applied. These protocols combine multiple low-quality entangled pairs to produce fewer pairs with higher fidelity [5, 16]. More advanced purification schemes and their performance characteristics have been studied in subsequent work, including generalized protocol families and analytical performance models [13, 23]. Importantly, different purification protocols exhibit distinct trade-offs between achievable fidelity improvement and success probability. Some protocols are more aggressive, achieving higher fidelity gains at the cost of increased failure risk, while others provide more conservative improvements with higher reliability.

However, purification is a probabilistic process and may fail, leading to the loss of all involved resources. As a result, there exists a fundamental trade-off between fidelity and availability in quantum communication systems [13, 19]. The existence of multiple purification protocols suggests that this trade-off can, in principle, be controlled more precisely by selecting appropriate strategies depending on the system state.

Alternatively, entanglement can be replaced by generating new pairs, which reduces the risk of complete resource loss but may not meet the fidelity requirements of quantum applications. These challenges highlight the need for intelligent and adaptive strategies that can dynamically balance fidelity and availability in quantum communication systems, while also selecting between different purification protocols to exploit their complementary properties.

1.2 PROBLEM STATEMENT AND CONTRIBUTION

In quantum communication systems, entangled links are consumable resources that must be managed efficiently to enable high-quality teleportation. Due to the stochastic nature of entanglement generation, decoherence, and probabilistic purification outcomes, this resource management problem becomes a challenging sequential decision-making task.

At each decision point, the system must choose between multiple actions: keeping the current entangled pair, replacing it with a newly generated one, or applying a purification protocol. Each of these actions introduces different trade-offs. Keeping an entangled pair preserves availability but may result in insufficient fidelity. Replacing the entanglement ensures a fresh resource but does not guarantee improved quality. Applying purification

can increase fidelity, but introduces the risk of failure, which may lead to the complete loss of entanglement and increased waiting times [13].

The problem is further complicated by stochastic application request arrivals and time-dependent decoherence effects, which influence both the availability and quality of entangled resources [11]. As a result, finding good decision policies analytically is very hard, as the probabilistic nature of the environment makes analysis difficult.

Existing work has explored the trade-off between fidelity and availability using analytical models [19], providing valuable insights into system behavior. Additionally, reinforcement learning has been applied to optimize teleportation fidelity in simplified settings, with a single purification protocol [24]. While these approaches demonstrate the potential of adaptive methods, they are limited in their ability to handle multiple purification strategies. Incorporating multiple protocols enables a more fine-grained control over the fidelity-availability trade-off. Different purification protocols exhibit distinct characteristics in terms of achievable fidelity improvement and success probability. As a result, some protocols favor aggressive fidelity enhancement at the cost of higher failure risk, while others provide more conservative improvements with higher reliability. A strategy based on a single purification protocol is inherently restricted to a fixed trade-off and cannot adapt its behavior to varying system states. In contrast, access to multiple protocols allows for state-dependent decision-making, where the selected action can be tailored to the current quality of the entanglement and system conditions.

Therefore, there remains a gap in designing adaptive strategies that can dynamically select between keeping, replacing, and multiple purification protocols under stochastic conditions.

To address this challenge, this thesis proposes a reinforcement learning-based framework that models the entanglement management problem as a Markov Decision Process. The framework learns adaptive policies that optimize a performance metric such as the average teleportation fidelity while accounting for stochastic system dynamics and application requirements. In contrast to static heuristics, the proposed approach enables flexible and data-driven decision-making in quantum communication systems.

The main contributions of this thesis are as follows:

- A detailed analysis of the performance of individual purification protocols under different error models and decoherence conditions.
- A reinforcement learning framework that models entanglement management as a Markov Decision Process and learns adaptive policies.
- The integration of multiple purification protocols into the decision space, enabling more flexible and effective strategies compared to single-protocol approaches.
- An extensive evaluation demonstrating the impact of adaptive policies with multiple purification protocols on average teleportation fidelity under varying system parameters.
- An extension of the framework to support optimize the average teleportation fidelity under varying wait time sensitivities.

BACKGROUND

2.1 QUANTUM COMMUNICATION

To understand Quantum Communication, we first need to understand how classical communication works. Classical communication sends bits e.g. 0 or 1. For example by light using optical fiber cables. This signal degrades over times, which makes amplifying this signal mandatory for larger distances. This is done by repeaters. These components read the signal and then retransmit it.

2.1.1 Qubits

In Quantum communication we don't have bits. There the basis is an analogous concept called qubits. A qubit is based on the two states $|0\rangle$ and $|1\rangle$. Using $|\rangle$ is called the Dirac Notation and is commonly used in quantum physics. It is the short notation for

$$|0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (2.1)$$

These two basis states form the computational basis of a qubit and correspond to the classical binary states 0 and 1, representing the fundamental measurement outcomes of a quantum system. Nevertheless, in contrast to a bit that can either be 0 or 1, a qubit can exist in an infinite number of states. The state of a qubit can be fully described by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (2.2)$$

where α and β are complex numbers and $|\alpha|^2 + |\beta|^2 = 1$ must be true. α and β are called the amplitude of the state ψ . A state $|\psi\rangle$ as defined above is called a pure state. But in quantum physics, we can also have mixed states. Fundamentally, a mixed state is a statistical ensemble comprised of a probabilistic mixture of pure states, denoted

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|. \quad (2.3)$$

where the described quantum system is in one of the states $|\psi_i\rangle$ with probability p_i . The fidelity F of a mixed state with respect to a wanted pure state measures how likely it is to receive the wanted pure state upon measuring the mixed state. [9, 26].

2.1.2 No Cloning Theorem And Measurement Postulate

A single qubit having the possibility to be in more states than a classical bit is a big advantage information that can be transported with one qubit. However, if we try to send a qubit e.g. as a photon over optical fiber, some of them get absorbed, meaning their state

is lost. For example, the best commercialized optical fiber has a attenuation of 0.15 db/km [12], which means that only 3,2 of 100 photons traveling a distance of 100km survive the way.

The absorption of qubits makes transmitting qubits over long distances a challenge. We cannot simply measure and resend them before they are lost. Measuring the qubits is not useful, due to the measurement postulate stating that if we measure a qubit, then it always give either $|0\rangle$ or $|1\rangle$, regardless of whether the qubit actually had values different to that [26]. Rather, $|\alpha|$ and $|\beta|$ determine the probabilities of obtaining one of the values, where α determines the likelihood of receiving $|0\rangle$ and β the likelihood of $|1\rangle$ [7]. Moreover, measuring a qubit alters its state. If it was in a superposition before, the state is lost and changed to the state that was measured. This process cannot be reverted [26].

We also can't clone the qubit, as it is impossible to create an exact copy of an arbitrary unknown quantum state [34]. This phenomenon is described by the no-cloning theorem. This prevents cloning the qubit multiple times and measuring the result, which would be a way to determine its state.

2.1.3 Multiple Qubits

In the quantum world we can use quantum repeaters to send qubits over longer distances [8]. These use a phenomenon called quantum entanglement. Qubits can be entangled, which means that the qubits become connected in way that upon measuring the first qubit, the state of the other qubit instantly fixes to the same state. So measuring the first qubit with 1, means that the second qubit will also yield 1 when measured [9]. Einstein called this "Spooky action at a distance". The most simple case of an entanglement would be having 2 qubits. In contrast to having two classical bits, that would have a state space of $\{00, 01, 10, 11\}$, 2 qubits can be described by a state space with the bases $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Analogous to a single qubit state, a two qubit state can be described by

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle \quad (2.4)$$

Again, the normalization condition $\sum_{x \in \{0,1\}^2} |\alpha_x|^2 = 1$ must hold true. The probability of a specific measurement outcome is determined by the coefficients $a_x \in \{0, 1\}$. Specifically, for any index $k \in \{00, 01, 10, 11\}$, the probability $P(|k\rangle)$ of measuring the system in the state $|k\rangle$ is given by $P(|k\rangle) = |\alpha_k|^2$ [26].

2.1.3.1 EPR-Pairs

An important form of a two qubit states are EPR-Pairs also called Bell-states: [9]

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2.5)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \quad (2.6)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (2.7)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \quad (2.8)$$

Each EPR pair exists as a superposition of the two basis states defined in its equation. Taking $|\Phi^+\rangle$ as an example, measuring the system will yield either the state $|00\rangle$ (both qubits being 0) or the state $|11\rangle$ (both qubits being 1). The probability of observing either state is exactly $\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$.

As a result, measuring the first qubit instantly determines the state of the second qubit. For instance, if the first qubit of the state $|\Phi^+\rangle$ is measured as 1, we immediately know that the second qubit must also be 1.

2.1.3.2 How to generate Entangled Qubits

Like in classical world, where we have gates like or and not, we have gates for qubits. E.g. CNOT Gate, Hadamard gate: [9]

Gate	Input Type	Transformation (Truth Table)	Effect
X-Gate	1 Qubit	$ 0\rangle \rightarrow 1\rangle, 1\rangle \rightarrow 0\rangle$	NOT-Operation
Z-Gate	1 Qubit	$ 0\rangle \rightarrow 0\rangle, 1\rangle \rightarrow - 1\rangle$	Phase-Flip
Hadamard	1 Qubit	$ 0\rangle \rightarrow \frac{ 0\rangle+ 1\rangle}{\sqrt{2}}, 1\rangle \rightarrow \frac{ 0\rangle- 1\rangle}{\sqrt{2}}$	Superposition
CNOT	2 Qubits (1st is control)	$ 00\rangle \rightarrow 00\rangle, 01\rangle \rightarrow 01\rangle$ $ 10\rangle \rightarrow 11\rangle, 11\rangle \rightarrow 10\rangle$	IF Control=1: Flip Target

Figure 2.1: Quantum gates and their properties

Assume that Alice and Bob each have a qubit that they want to entangle. The two qubits can be represented by $|00\rangle$, where the Alice qubit is represented by the first 0 and the Bobs qubit represented by the second 0. To start the entanglement process, Alice applies a hadamard gate to her qubit, resulting in her qubit having the state

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (2.9)$$

As shown in the table, the hadamard gate changes the qubits state into a superposition. The two qubit state now is

$$\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle). \quad (2.10)$$

Till now the qubits are not entangled. To achieve the entanglement a CNOT gate is applied to the two qubit state, where Alice qubit is the control qubit and Bobs is the target. If Alice qubit is $|0\rangle$ then Bobs qubit stays $|0\rangle$. If Alice qubit is $|1\rangle$, then Bobs qubit becomes $|1\rangle$. The result is the Bell-State [22]

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (2.11)$$

We do not know what Alice and Bobs state are, who only know that they are the same when measured. Thus the two qubits are now entangled, because measuring one qubit instantly fixes the value of the other qubit to the same value [9].

We can not just measure our part of the entanglement, because if we do the state is altered and we cannot use it for further operations that need the original super position state. To actually send over quantum information in superposition state, we need quantum teleportation.

2.1.4 Quantum Teleportation

Quantum teleportation is the process of transmitting a qubit without the physical transfer of the particle storing the qubit [9]. It uses an entanglement as a resource to transfer a qubit with the aid of classical communication. Assume Alice and Bob each have a qubit of the EPR-Pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Alice also has a qubit $|\Phi\rangle$ that she wants to send over to Bob. The global state can be described by

$$|\varphi_1\rangle = |\Phi\rangle \otimes |\Phi^+\rangle = \alpha |0\rangle \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} + \beta |1\rangle \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (2.12)$$

The operator $\otimes$ denotes the tensor product and is used to combine the states of multiple quantum systems into a joint system. Given two quantum states $|\psi\rangle$ and $|\phi\rangle$, the state of the composite system is defined as [9]

$$|\psi\rangle \otimes |\phi\rangle.$$

When following the convention that the two leftmost qubits belong to Alice and the rightmost qubit belongs to Bob, we can simplify the formula to

$$|\varphi_1\rangle = (\alpha |000\rangle + \alpha |011\rangle + \beta |100\rangle + \beta |111\rangle) / \sqrt{2}. \quad (2.13)$$

The first step of the teleportation process is Alice applying the CNOT gate to her pair of qubits which maps the state $|10\rangle$ to $|11\rangle$ and vice versa, which results in the global state

$$|\varphi_2\rangle = (\alpha |000\rangle + \alpha |011\rangle + \beta |110\rangle + \beta |101\rangle) / \sqrt{2}. \quad (2.14)$$

Then Alice applies the Hadamard gate on the qubit to be teleported which alters a qubits state to be in a superposition. The resulting global state is

$$|\varphi_3\rangle = (\alpha |000\rangle + \alpha |100\rangle + \alpha |011\rangle + \alpha |111\rangle + \beta |010\rangle - \beta |110\rangle + \beta |001\rangle - \beta |101\rangle) / 2 \quad (2.15)$$

As a third step, Alice measures the two qubits on her side. This measurement has four possible outcomes with equal probability of 25%: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The result of the

Alice outcome	Bob's state	Instruction to Bob	Result
$ 00\rangle$	$\alpha_0 0\rangle + \alpha_1 1\rangle$	None	$\alpha_0 0\rangle + \alpha_1 1\rangle$
$ 01\rangle$	$\alpha_0 0\rangle - \alpha_1 1\rangle$	Z	$\alpha_0 0\rangle + \alpha_1 1\rangle$
$ 10\rangle$	$\alpha_1 0\rangle + \alpha_0 1\rangle$	X	$\alpha_0 0\rangle + \alpha_1 1\rangle$
$ 11\rangle$	$-\alpha_1 0\rangle + \alpha_0 1\rangle$	X then Z	$\alpha_0 0\rangle + \alpha_1 1\rangle$

Table 2.1: Operations Bob needs to apply

measurement is sent over to Bob via classical communication. Upon measuring her qubits, Bob's qubit is fixed to a state which depends on the result Alice got.

The fourth step is Bob applying local operations on his qubit, depending on the operations received from Alice. The operations are shown in Table 2.1 [9, 22]

As shown quantum teleportation can be used to transit a qubit's state from repeater to repeater. Using this technique we can form larger networks that work analogous to the classic internet [4]. Importantly, after using an entanglement to teleport a qubit, the entanglement is lost and no longer available for teleportation. Thus for each teleportation, one entanglement is needed.

2.1.4.1 Noise

The teleportation example above assumed perfect conditions. However, in reality we have noise in the quantum world, which exponentially degrades the quality of single qubits and entanglements over time. This process is called decoherence and the speed of decoherence is classified by the decoherence rate t_c [11].

Single Qubit fidelity decays due to dephasing noise which describes the loss of energy to the environment [26]. The fidelity of a single qubit after spending some time in memory is described by [11]

$$F(t) = |\alpha|^4 + 2e^{-\Gamma t}|\alpha|^2|\beta|^2 + |\beta|^4 \quad (2.16)$$

The fidelity of entangled qubits degrades over time due to environmental noise. Because of the noise the two qubit state does not stay in the target pure state $|\Phi^+\rangle$. Over time, the noise can bring the two qubit state in one of the possible four Bell-States with some probability. Formally, this is called Bell-Diagonal State which is described by [23]

$$\rho_w = F|\Phi^+\rangle\langle\Phi^+| + \lambda_1|\Psi^+\rangle\langle\Psi^+| + \lambda_2|\Psi^-\rangle\langle\Psi^-| + \lambda_3|\Phi^-\rangle\langle\Phi^-| \quad (2.17)$$

where the coefficients F and λ_{1-3} sum up to 1 and represent the probabilities associated with their respective states. $|\Phi^+\rangle$ is the target state. Thus F denotes the fidelity of a Bell-state.

If $\lambda_1 = \lambda_2 = \lambda_3$ then the Bell-State is in a special form, called Werner-State. This state can be fully described by its fidelity F . When applying extra noise any Bell-state can be transformed to a Werner state by a process called twirling [PhysRevA.54.3824].

The decoherence process of entangled qubits is remarkably complex; even for relatively simple systems consisting of only three qubits, an arbitrary noise process is determined by approximately 4,000 independent real parameters [18]. Fortunately, Dür et al. demonstrated

that by applying correlated, random local unitary operations, which is a procedure known as depolarization or 'twirling', the parameters can be drastically reduced to a simplified standard form. In many practical scenarios, the noise can be further reduced to a single parameter, representing global white noise. Consequently, the decay of entanglement fidelity F_e can be modeled as [18]

$$F_e(t) = e^{-(t-t_0)/t_c}(F_e(t_0) - 0.25) + 0.25 \quad (2.18)$$

Interpreting the equation one can see the worst fidelity an entangled pair can have is $\frac{1}{4}$. This is the case, when all states are equally likely and the pair doesn't carry any information.

Additionally, quantum gates are not perfect. Creating an entanglement as shown in [Section 2.1.3.2](#) doesn't necessarily create an entanglement with perfect fidelity of 1.

2.1.5 Performance Measurement

Applications such as Quantum Key Distribution and distributed quantum applications relying on quantum teleportation work better with higher fidelity entanglements. Some applications are also strongly sensitive to wait times. Thus the performance of a quantum network is determined by the metrics fidelity of entanglement and the availability. The fidelity is crucial, because it determines the fidelity of the teleported qubit that is used for the application. The waiting time is important, because when an application qubit is waiting for an entanglement to be available it loses its fidelity. The importance of the waiting time is depended on the coherence time, because a smaller time results in a faster loss of fidelity for the application qubit. Also, some applications are more sensitive to waiting times than others.

To fight the degrading quality of the entanglement a strategy could be to always replace the entanglement by a new one, once the fidelity becomes too bad. Although this works, there is a process that in some cases gives even better results. The process is called purification.

2.1.6 Purification

Improvement of the performance can be achieved by quantum purification, quantum error detection and error correction [4]. In this work we will focus on using purification. Purification is a process where two separated observers can use multiple entanglements to create a lesser number of entanglements with higher fidelity by applying local operations [5]. The creation of such higher fidelity links seems very useful, however purification comes at a risk: The process can fail, which leads to losing all initial entanglements and not creating new ones. This results in lower availability, because we have no entanglements to teleport qubits.

The first purification protocol was given by Bennett et al. [6]. They proposed the BBPSSW protocol. It works on identical copies of a Werner-state and purifies these consuming one of the entanglements. This process is repeated multiple times. Protocols working in this fashion are grouped by the name recurrence schemes. A second one is called DEJMPS [16] which works in a similar way, but takes better advantage of the information given by the entanglements than BBPSSW and thus improves the entanglements faster. Additionally,

DEJMPS can work on all Bell-diagonal states, not only Werner-states.

Because each step of a recurrence scheme consumes two identical pairs of a specific fidelity to produce one pair of a higher fidelity, the number of initial raw states required scales exponentially with the number of purification rounds. A purification scheme giving a solution to this problem was proposed by Dür et al. [17]. The scheme is called pumping and is based on having a base entanglement in some memory, that is repeatedly improved by applying purification on it and repeatedly generated auxiliary entanglements. For the actual purification it uses the same purification map as BBPSSW and DEJMPS.

In Jansen et al. [23], a family of purification protocols known as bilocal Clifford protocols was introduced. Protocols operating on Bell-diagonal states belong to this class if they employ arbitrary noiseless bilocal Clifford circuits, retain a single qubit pair, measure the remaining pairs, and allow Alice and Bob to exchange the measurement outcomes.

Among this family, three non-trivial bilocal Clifford protocols are commonly considered. While the full family contains additional variants, these are considered trivial, as they perform worse than simply replacing or keeping existing entanglement [13].

In this work, we consider four different pumping protocols. Three of them are selected from the bilocal Clifford family, originally introduced by Dehaene et al. [14] and later reformulated into a practically usable form by Davies et al. [13]. The fourth protocol is Polarization Mode Dispersion (PMD) [27]. In general, purification protocols can be characterized by their success probability and the fidelity of the resulting entangled pair. Furthermore, they differ in the types of errors they are able to mitigate.

Bilocal Clifford protocols are particularly attractive due to their relatively simple implementation. They can be realized using bilocal Clifford operations followed by a single round of classical communication, making them promising candidates for near-term experimental implementations [23]. The three considered protocols target different error components. However, for Werner states, all three protocols yield identical results in terms of both success probability and resulting fidelity.

In contrast, the PMD protocol is more restrictive, as it is only applicable when $\lambda_2 = \lambda_3 = 0$. This corresponds to entanglement being in either the Bell state $|\Phi^+\rangle$ or $|\Phi^-\rangle$.

The control of such noisy quantum systems is very complex analytically. A learning based approach seems useful to still find a solution.

2.2 DEEP REINFORCEMENT LEARNING

We want to create a policy π that uses purification optimally. That is choosing the right action given a state, where the actions are keeping the best entanglement or using one of the Purification Protocols. Finding such a policy is highly complex in the quantum world, as a lot of things only happen with a certain success probability, such as the entanglement generation, qubit arrival and purification. This makes the quantum world a probabilistic environment, making analytical analysis really hard. However, such probabilistic environments can be efficiently modeled as a Markov Decision Process(MDP). A MDP is a five-tuple of [1, 30]

1. Set of states $\mathcal{S}$

2. Set of actions $\mathcal{A}$
3. Transition function $P(s_t|a_t, s_{t+1})$, denoting the probability of transitioning to state s_{t+1} when starting in state s_t and taking action a_t
4. Reward function $T(a_t, s_t, s_{t+1})$, denoting the reward of transitioning from state s_t to s_{t+1} by choosing action a

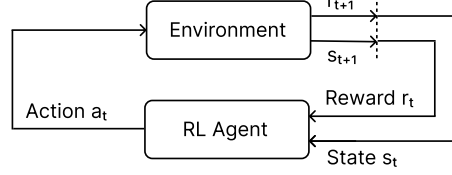


Figure 2.2: Agent-Environment-Interface

MDPs can be solved using Reinforcement Learning. Reinforcement Learning is about an agent interacting with an environment, represented by a state s . The agent can observe the current state s_t . Based on that he can take an action a_t . The chosen action leads to a change in the environment, which in turn emits a new state and a reward. Based on the new state and reward the agent can take a next action. This loop is often called the Agent-Environment-Interface.

The goal of the agent is to maximize the cumulative reward over time. For an agent starting at time-step t , the return G_t is the total discounted reward from that time onward:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (2.19)$$

where R_{t+k+1} is the reward received after action at time $t+k$ and $\gamma \in [0, 1]$ is the discount factor that determines the importance of future rewards[30].

During interacting with the environment the agent creates a trajectory $\tau = (s_0, a_0, s_1, a_1, \dots)$, which is a sequence of states and actions [1]. An important function in Reinforcement Learning is the value function. RL-Algorithms use it to estimate the value of being in a state when applying a given policy from that state onward.

$$v_{\pi}(s) = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \middle| S_t = s \right] \quad (2.20)$$

In general, there are two paradigms in reinforcement learning: model-free methods and model based methods. The first learns directly from experience, without explicitly modeling the environment meaning they don't try to model the transition- and reward-function of the MDP. The model-based methods try to learn a model of the environment and uses it for planning. A special type of model-free methods is policy optimization that directly learn the policy $\pi_{\theta}(a|s)$. These techniques generally improve performance by adjusting parameters θ through gradient ascent on the objective function $J(\pi_{\theta})$, or alternatively, by maximizing local approximations of that objective. These methods are typically characterized by an on-policy approach, where parameter updates rely exclusively on data generated by the current policy iteration. Furthermore, this process frequently incorporates a learned approximator for the on-policy value function, which serves as a baseline or guide for the policy refinement[1].

2.2.1 Policy Gradient Methods

Policy gradient methods directly learn the policy $\pi_\theta(a|s)$. These techniques optimize the parameters θ of a differentiable policy by performing gradient ascent on an estimate of the expected return $\nabla_\theta J(\theta)$. The objective function $J(\theta)$ represents the expected cumulative discounted reward over a trajectory τ and is defined as:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \right]$$

In practice, vanilla policy gradients frequently suffer from high variance, which can lead to unstable learning processes. To mitigate this issue, variance reduction techniques are employed, such as the introduction of a baseline typically a learned value function $V(s)$ —and advantage estimates $A(s, a)$.

This refinement yields the actor-critic architecture. The actor represents the policy π_θ and is responsible for selecting actions. The critic estimates the value function $V(s)$, providing feedback to the actor to guide the policy update and reduce the variance of the gradient estimate.

2.2.2 Neural Networks

For simple problems, we can learn the best action for each state and save that in a table, as is done in Tabular Q-Learning [33]. The table can then be consulted, at each time step to get the best action. But this is not possible most of the times as the state spaces are too large to save all state-action pairs independently. To solve this problem neural networks are often used as a policy and/or value-function [3]. A neural network is characterized by a succession of multiple processing layers. It has an input layer, which receives the input values. These are then passed onto the next layer(s). Doing that, the values are transformed with a non-linear function, parameterized by some constant called weight. [31]

2.2.3 Proximal Policy Optimization

Proximal Policy Optimization (PPO) is a family of policy gradient methods. In contrast to the more simpler policy gradient methods they don't use a standard policy gradient but optimize a surrogate function using gradient ascent. The intention behind that is to prevent large policy updates to cause performance collapse of the policy. PPO methods are model-free and use an actor-critic. This means they let the policy (actor) learn by using feedback from the value function (critic) [3]. PPO can use each batch of on-policy data for several update-steps, which improves sample efficiency [28]. The Pseudo code of the PPO-Clip, can be found in 1.

Algorithm 1 Pseudo code of PPO-Clip [1, 29]

- 1: Input: initial policy parameters θ_0 , initial value function parameters ϕ_0
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: Collect set of trajectories $\mathcal{D}_k = \{\tau_i\}$ by running policy $\pi_k = \pi(\theta_k)$ in the environment.
- 4: Compute rewards-to-go $\hat{R}_t$.
- 5: Compute advantage estimates, $\hat{A}_t$ (using any method of advantage estimation) based on the current value function V_{ϕ_k} .
- 6: Update the policy by maximizing the PPO-Clip objective:

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \min \left(\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)} A^{\pi_{\theta_k}}(s_t, a_t), \ g(\epsilon, A^{\pi_{\theta_k}}(s_t, a_t)) \right),$$

typically via stochastic gradient ascent with Adam.

- 7: Fit value function by regression on mean-squared error:

$$\phi_{k+1} = \arg \min_{\phi} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T (V_{\phi}(s_t) - \hat{R}_t)^2,$$

typically via some gradient descent algorithm.

- 8: **end for**
-

RELATED WORK

This thesis studies adaptive entanglement management at the link level in quantum communication systems, motivated by decoherence, probabilistic entanglement generation, and the risk–reward trade-off imposed by entanglement purification. We structure related work into three sections: (i) entanglement purification and distillation protocols and their modeling, (ii) link-layer buffering and scheduling under limited quantum memories, and (iii) adaptive control and learning-based approaches for quantum-network resource decisions.

3.1 ENTANGLEMENT PURIFICATION AND DISTILLATION

Early recurrence-style entanglement purification protocols established that local operations and classical communication can transform multiple noisy Bell pairs into fewer higher-quality pairs, forming a foundation for faithful teleportation and secure communication over noisy channels [5, 16]. Quantum-repeater architectures combined purification on short segments with entanglement swapping to overcome loss and noise at long distances, while highlighting the sensitivity to imperfect local operations [17]. Beyond these foundational results, recent work has systematically explored the design space of purification circuits and protocol families. Automated search methods produce short circuits with improved success probability and output fidelity under realistic error models [25], and symmetry-based enumeration provides optimized bilocal Clifford distillation protocols for Bell-diagonal states [13, 23]. Other approaches exist. E.g. [27] proposes a purification protocol that uses properties only existing when using optical fiber. These works are valuable for defining the menu of candidate purification actions available to an adaptive controller. In contrast to this, our work does not try to improve a single protocol, but to better decide when to apply which protocol and which protocol to use in one setting.

3.2 LEARNING-BASED CONTROL

In [24], reinforcement learning is employed to improve the average teleportation fidelity in a two-node, two-memory quantum network. The RL agent is given the binary choice of applying purification (pumping) or not, using a single pumping protocol proposed in [27]. This approach proves effective, as it achieves higher expected teleportation fidelity across different decoherence times compared to the baseline strategies of always pumping and never pumping. Furthermore, the learned strategy is able to outperform these baselines under varying waiting time sensitivities. However, since only a single pumping protocol is considered, there remains significant potential for further improvement.

In this work, we adopt a similar reinforcement learning framework but extend the action space to include multiple pumping protocols, which is expected to yield additional performance gains.

A lot more approaches exist when it comes to using machine learning (ML) in quantum communication as shown in [32]. They used machine learning to find teleportation protocols

while minimizing the amount of operations used, given a set of gates. A similar setup was used to find purification protocols. The system could find a protocol similar to DEJMPS protocol [16]. Additionally, they used ML to improve on the idea of quantum repeater[8]. In contrast to our work, they did not use two nodes, but three, where the node in the middle was shared between the other two nodes. The agent was given the option to use the protocols for entanglement teleportation, entanglement swapping and entanglement purification discovered before to find a protocol for distributing an entangled state between the two outer stations with a threshold fidelity of at least 0.9, all the while using as few initial states as possible. This contrasts to our work, where we do not try to reach a certain threshold, but to find the best fidelity achievable given the decoherence time.

4.1 MODEL

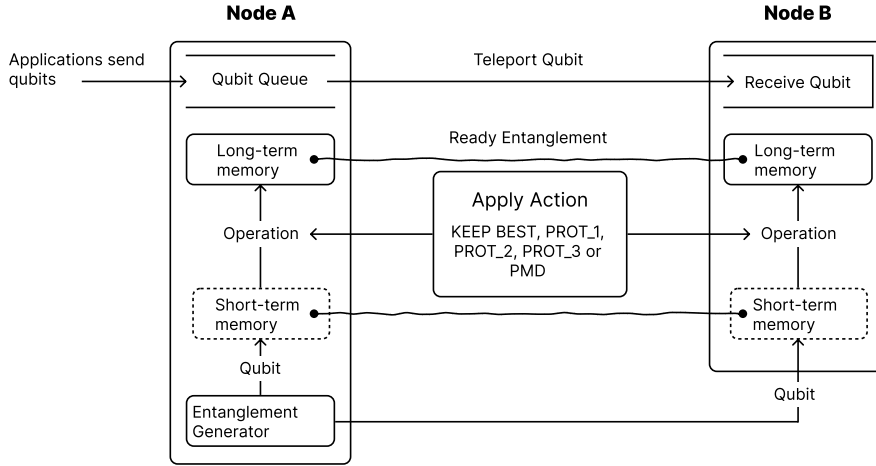


Figure 4.1: System Model

In this work, we consider a quantum communication system consisting of two nodes, denoted as Node A and Node B, between which quantum information is transmitted via quantum teleportation. The system relies on entangled qubit pairs as a consumable resource that are continuously generated.

1. **Nodes and Memory Architecture:** Each node is equipped with two quantum memories. A short-term memory in which newly generated entanglements are stored in and a long-term memory, which stores the entanglement used for teleportation. Entangled pairs are modeled as Bell-diagonal states of the form

$$\rho_s = F_E |\phi^+\rangle\langle\phi^+| + \lambda_1 |\psi^+\rangle\langle\psi^+| + \lambda_2 |\psi^-\rangle\langle\psi^-| + \lambda_3 |\phi^-\rangle\langle\phi^-| \quad (4.1)$$

where F_E denotes the fidelity with respect to the target Bell state $|\phi^+\rangle$, and $\lambda_1, \lambda_2, \lambda_3$ represent error contributions. Entanglement E will be denoted $E(\lambda_1, \lambda_2, \lambda_3)$ for short, where $F_E = \lambda_1 - \lambda_2 - \lambda_3$ is the fidelity of the entanglement E . Newly generated entanglements are first placed in short-term memory. We assume that they spent no time there but need to be instantly used in one of the following ways: they can be discarded, used for purification, or replace the entanglement in long-term memory. The entanglement stored in long-term memory is assumed to be in the Werner-state:

$$\rho_l = F_l |\phi^+\rangle\langle\phi^+| + \frac{1-F_e}{3} |\psi^+\rangle\langle\psi^+| + \frac{1-F_e}{3} |\psi^-\rangle\langle\psi^-| + \frac{1-F_e}{3} |\phi^-\rangle\langle\phi^-| \quad (4.2)$$

Thus, before entanglements are saved in long-term memory they must be transformed from a general Bell-state into a Werner-state by applying twirling. The fidelity of the entanglement in long-term memory is termed F_l . The nodes are connected by an optical fiber cable that is used for classical communication. The cable is defined by its length l and attenuation η , which describes how likely it is that photons send over the cable are absorbed.

2. **Entanglement Generation:** We assume that entanglement generation happens in Node A. After generation, one of the entanglement qubits need to be sent to Node B as a photon via classical communication. When being send, the qubit can be absorbed with probability $e^{-\eta l}$, which leads to the entanglement not being available. We model this process as a Bernoulli process with success probability $p_g = 1 - e^{-\eta l}$, denoting the probability that the photon is not absorbed. Generation takes time Δt_g , which models the time to transfer the needed information between the two nodes according to $\frac{l}{c}$. The generated entanglement has fidelity F_0 . The newly generated entanglement is stored in short-term memory.
3. **Decoherence and Fidelity Evolution:** As a result of decoherence the fidelity of the entanglement stored in long-term memory decreases over time. In this work, we assume depolarizing noise as the dominant error source for entanglements, since it models the most significant type of quantum noise [18]. Under this model, the entanglement fidelity evolves according to

$$F_l(t) = e^{-\frac{(t-t_0)}{t_c}} (F_e(t_0) - 0.25) + 0.25, \quad (4.3)$$

where t_0 is the creation time of the entanglement and t_c is the coherence time. This is the same assumption [13, 24] use. Note that the noise is isotropic, thus affecting all the parts of the bell-state equally. Single qubits in memory also decay. We consider only dephasing noise as a factor. The average decay of the single-qubit fidelity is assumed to follow [24]

$$\bar{F}_q(t, \kappa) = (e^{-\frac{(t-t_0)\kappa}{t_c}} + 2)/3 \quad (4.4)$$

where t_0 is the creation time of the qubit, t_c the coherence time and κ controls the wait time sensitivity. κ is used to control how sensitive the application sending the qubit is to waiting for a qubit to be teleported. In both cases t_c determines how fast the fidelity of the entanglements or qubits are degraded.

4. **Purification:** To counteract decoherence, purification protocols are applied on the entanglement in long-term memory described by F_l using the entanglement $E(\lambda_1, \lambda_2, \lambda_3)$ from short-term memory. In this work, multiple pumping protocols are considered, each characterized by a jump function, denoting the resulting fidelity of the entanglement in long-term memory, when successful and the success probability, which calculates how likely it is that the purification succeeds. The resulting fidelity depends on F_l , F_E and the error distribution of E given by λ_{1-3} . The protocols used in this work are taken from [13, 27]

Purification is assumed to take negligible time compared to entanglement generation.

Protocol	Jump function	Success probability
PROT ₁	$\frac{(4\lambda_1+3\lambda_2+3\lambda_3-3)F_e-\lambda_1}{(4\lambda_2+4\lambda_3-2)F_e-\lambda_2-\lambda_3-1}$	$\frac{2}{3}(1-2\lambda_2-2\lambda_3)F_e + \frac{1}{3}(1+\lambda_2+\lambda_3)$
PROT ₂	$\frac{(3\lambda_1+4\lambda_2+3\lambda_3-3)F_e-\lambda_2}{(4\lambda_1+4\lambda_3-2)F_e-\lambda_1-\lambda_3-1}$	$\frac{2}{3}(1-2\lambda_3-2\lambda_1)F_e + \frac{1}{3}(1+\lambda_3+\lambda_1)$
PROT ₃	$\frac{(3\lambda_1+3\lambda_2+4\lambda_3-3)F_e-\lambda_3}{(4\lambda_1+4\lambda_2-2)F_e-\lambda_1-\lambda_2-1}$	$\frac{2}{3}(1-2\lambda_1-2\lambda_2)F_e + \frac{1}{3}(1+\lambda_1+\lambda_2)$
PMD	$\frac{F_E F_I}{F_E F_I + (1-F_E)(1-F_I)}$	$F_E F_I + (1-F_E)(1-F_I)$

Table 4.1: Overview of the pumping protocols with their jump functions and success probabilities.

5. **Qubit Queue and Teleportation:** Application qubits arrive at Node A according to a Poisson process with rate λ_q . Each qubit has initial fidelity $F_q = 1$ which degrades while waiting in the queue. If an entanglement is available in long-term memory, teleportation is performed immediately. Otherwise, the qubit waits until an entanglement becomes available. Teleportation consumes the entangled pair. The resulting teleportation fidelity depends on both entanglement and qubit fidelity and is given by: [24]

$$\bar{F}_T(F_I) = \frac{2F_I + 1}{3} \quad (4.5)$$

With noisy qubits the average fidelity of teleportation given the average fidelity and the entanglement fidelity can be described in a simplified way by [24]

$$\bar{F}_T(F_I, \bar{F}_q) = \left(\sqrt{\frac{(2F_I + 1)\bar{F}_q}{3}} - \sqrt{\frac{2(1-F_I)(1-\bar{F}_q)}{3}} \right)^2 \quad (4.6)$$

The system operates dynamically, driven by two asynchronous stochastic processes: the continuous generation of entangled pairs and the arrival of application qubits. These parallel processes dictate the lifecycle and availability of the network's resources as shown in Figure 4.2.

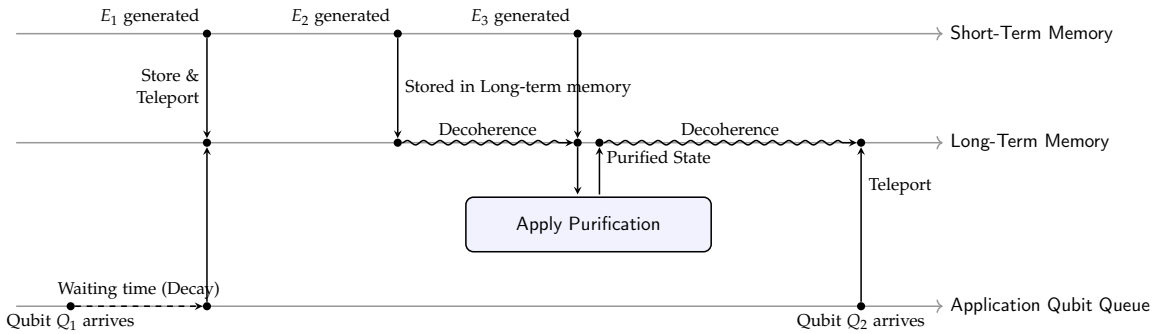


Figure 4.2: Timeline of asynchronous system events. Application qubits and entanglements arrive independently. The system can perform different operations given an entanglement in short-term and long-term memory. In this example, purification was applied.

Newly generated entangled pairs first enter the short-term memory. If the long-term memory is available, they are transferred immediately to serve as ready resources. However, if the long-term memory is already occupied, a resource conflict occurs. To resolve this, the

system must execute a operation. The available operations encompass all possible physical actions: retaining the best available pair (KEEP BEST), or attempting to combine resources through purification using either PROT_1, PROT_2, PROT_3, or PMD.

Independently, application qubits arrive and demand teleportation. If an entanglement is ready in the long-term memory, it is consumed instantly to teleport the qubit. If not, the application qubit is buffered in a queue. During this waiting period, its fidelity inevitably degrades due to decoherence until a new entanglement is established. The overarching challenge of the system is therefore balancing the proactive improvement of entanglement fidelity against the risk of leaving arriving qubits waiting without usable resources.

4.2 PROBLEM DEFINITION AND APPROACH

The objective of our system is to maximize the expected utility, which is defined as the average teleportation fidelity of an application qubit. As introduced, this utility U depends on the fidelity of the entanglement in long-term memory, the waiting time of the qubit, and the wait time sensitivity:

$$U = \mathbb{E}[\bar{F}_T(F_l, \bar{F}_q(t, \kappa))] \quad (4.7)$$

This utility is also used in [24]. To systematically optimize this utility, we define a policy, denoted as π . A policy defines the behavior of the system at any given decision point by mapping the current state $s \in \mathcal{S}$ to a probability distribution over the available actions $a \in \mathcal{A}$. The policy accounts for all possible actions, including retaining the best entanglement (KEEP BEST), and the purification protocols PROT_1, PROT_2, PROT_3, and PMD:

$$\pi(a|s) = \mathbb{P}[A_t = a | S_t = s] \quad (4.8)$$

Our goal is to find the optimal policy π^* that yields the highest possible expected utility:

$$\pi^* = \arg \max_{\pi} U(\pi) \quad (4.9)$$

Finding this optimal policy analytically is intractable due to the highly stochastic nature of the quantum environment, which includes probabilistic entanglement generation, uncertain purification outcomes, and random application qubit arrivals. Therefore, we approach this challenge by modeling the system as a Markov Decision Process (MDP) and employing Reinforcement Learning (RL) to find a learned approximation of the optimal policy.

4.2.1 Modeling As Markov Decision Process

The Markov Decision Process is defined as follows: The state is given by

$$S = (F_l, t_r, q, \lambda_1, \lambda_2, \lambda_3) \in [0.25, 1] \times \mathbb{R} \times \{0, 1\} \times [0, 1]^3 \quad (4.10)$$

F_x represents the fidelity of the memory in long-term memory which is in a Werner-state and thus can be completely represented by the fidelity. Storing this fidelity in the state is important, as it is the main influence for the teleportation fidelity of the application qubit. t_r describes the time since the last application qubit arrived. The inclusion of t_r is fundamental for managing the trade-off between fidelity and availability, as it serves as a proxy for

the imminent demand on the network's resources. Due to the Poisson arrival process, the probability of a new request occurring in any infinitesimal time interval remains constant; however, from the agent's perspective, t_r defines the window of opportunity for resource improvement. As t_r increases, the likelihood of a qubit arrival in the immediate future grows, which shifts the cost-benefit analysis of purification. While a successful purification would increase fidelity, the associated risk of failure becomes increasingly unacceptable when a request is likely to arrive. In such states, the penalty for a missed immediate teleportation outweighs the possible fidelity gain of purification. q can either be 0 meaning no application qubit is waiting in the queue to be teleported or 1 in case a qubit is waiting for teleportation. λ_{1-3} each represent the lambdas of the newly generated entanglement that is in a general Bell-Diagonal state and stored in short-term memory. The agent needs to know about the lambda values, as the different pumping protocols yield different results depending on the error distribution given by the lambda values.

The possible actions are KEEP BEST, and the purification protocols PROT_1, PROT_2, PROT_3 and PMD given in Table 4.1. KEEP BEST means comparing the fidelity of the entanglement in long term memory to the fidelity of the link in short-term memory and keeping the best of the two. If the link in long-term memory has higher fidelity it is kept there and newly generated link is discarded. Otherwise the link in long-term memory is discarded and the newly generated link is placed there. This is needed for multiple reasons. For low coherence times, entanglements degrade so fast that purification is either not possible entirely or using it does not yield better results than keeping the newly generated entanglement [24]. Moreover, for higher coherence times, KEEP BEST place a crucial roll in controlling the fidelity-availability trade-off. As keeping the best entanglement does not have a risk of losing the entanglement in long-term memory, but also does not improve on the fidelity beyond what is available by the environment. Using it results in high availability, but sub-optimal fidelity. This is how we lean in on one side of the trade-off. The fidelity-optimization side is given by applying one of the other four actions, which result in the agent applying the respective purification protocol. This can result in a higher fidelity, but also bears the risk of losing entanglements, which lowers the availability. The combination of both types of actions is what lets us control the trade-off. In contrast to existing work [24], we give our model the option to use different purification protocols, each having different jump functions and success probabilities. We assume that this approach yields better results than only using a single protocol. The reasoning behind this is that having different protocols, each having different properties gives us an advantage in some situations where only having one protocol is imperfect. Additionally, PROT_1-3 can be applied for all Bell-Diagonal States and PMD can only be used for Bell-Diagonal States with $\lambda_{2,3}$ equal to 0.

The transition function $P(s_t|a_t, s_{t+1})$ describes how the MDPs state transitions from the current state to the next one, given the agents action. One such interaction is a step. The sequence of steps from start to finish are called episode. The transition function is based on a simulation of the system model. It includes all the assumptions presented in Section 4.1. An episode begins in the start state, which triggers an event generation. This can either be an entanglement generation or a application qubit arrival. In response to an entanglement generation, which generates an entanglement in short-term memory, the system checks

wether the long-term memory is empty. If true, then the entanglement is moved there and a new event is wait out. If the long-term memory is not empty, then the system needs an action from the agent. This is where one step ends. The state is set to represent the system state. Given the state, the agent decides on an action that will be applied to the environment in the next step. After that the next event is wait out. Notably, multiple events can happen in one step. The transition from one state to another only happens whenever agent input is needed or the episode ends. If the event is a qubit arrival, the system checks whether an entanglement in long-term memory exists. If not, the qubit is not teleported and a new event is wait out. If the step happens to end when a qubit is waiting, the property q of the state is set to 1, otherwise it is always set to 0. If an entanglement exists in long-term memory, the qubit is teleported and the episode ends. The teleportation of the qubit can also be triggered, without a qubit arrival event happening, if a qubit already waiting in the queue and a entanglement in long-term memory arrives.

The reward function $R(S_t, A_t, S_{t+1})$ defines the feedback the agent receives. To align the RL agent's objective with our system's utility U , we design a sparse reward structure. A non-zero reward is exclusively granted at the terminal step of an episode where a qubit is teleported. The reward is set to the teleportation fidelity upon serving a qubit. For all intermediate steps, the reward is strictly 0:

$$R(S_t, A_t, S_{t+1}) = \begin{cases} \bar{F}_T(F_l, \bar{F}_q(t, \kappa)) & \text{if qubit is teleported} \\ 0 & \text{else} \end{cases} \quad (4.11)$$

In Reinforcement Learning, algorithms seek to maximize the expected cumulative discounted reward, known as the expected return $J(\pi)$ [1]:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t, S_{t+1}) \right] \quad (4.12)$$

By setting the discount factor to $\gamma = 1.0$ and utilizing our sparse reward function, the sum of rewards for any successful trajectory collapses to exactly the final teleportation fidelity. Consequently, the expected return optimized by the RL agent maps exactly to our utility function:

$$J(\pi) \equiv U(\pi) \quad (4.13)$$

This equivalence guarantees that optimizing the MDP directly optimizes our utility. [Figure 4.3](#) highlights the particularities of our approach.

4.2.2 Using Proximal Policy Optimization

We use Proximal Policy Optimization (PPO) to find the optimal policy π_* defined in [Equation 4.9](#). PPO is chosen, because it has a good balance between sample complexity, simplicity, and wall-time [28]. Additionally, it works with discrete action spaces and has shown to be successful in similar settings [24], as well as in more complex settings [21]. Each, policy and value-function have a dedicated neural network with two hidden layers with 256 nodes each and tanh activation function. The other hyperparameter of PPO will be found using a Hyperparameter tuning framework, which is given a state parameter space in which it tries to find the best parameters by running multiple trials with different parameters sampled from the parameter space.

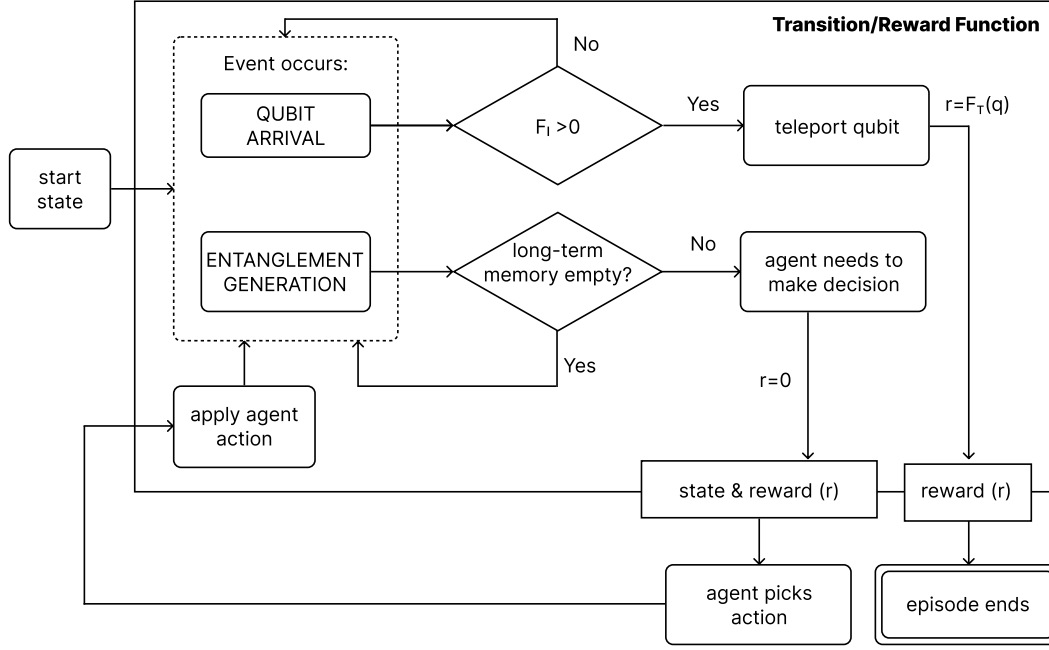


Figure 4.3: The agent interacting with the MPD (simplified, does not show all things that can happen). Points out that multiple events can occur in one transition from one step to another, without the agent seeing the event-loop from the outside. The agent only sees the environment through the state and reward, whenever the environment needs a decision from the agent, not whenever an event happens. Also shows that reward is only other than 0 if a qubit is teleported and then, the episode ends instantly.

4.3 IMPLEMENTATION

4.3.1 Environment

To implement the MDP, the python library Stable-Baselines 3 library [29] is used. To apply reinforcement learning, first the creation of a training environment is needed, which is based on our MDP. The Environment contains the functions `init`, `reset` and `step` that need to be implemented.

The `init` function creates a observation space, which is equal to the state of the MDP with the difference that all six parameters are set to range between $[0, 1]$ based on the idea that uniform state spaces makes it easier for the neural network to learn.

The `reset` function is called directly after the `init` function is called and whenever an episode ends. The `reset` function advances the simulation until the agent has a decision to make. This is the case whenever the event `ENTANGLEMENT_GENERATION` happened and the long-term memory is already occupied. Because if the long-term memory is empty and an entanglement arrives it can directly be placed there. And if the event is `QUBIT_ARRIVAL`, the qubit can just be teleported.

The `step` function is called by the agent after it observed the environment. Calling the function the agent passes his chosen action based on the observed state. This action is then processed by the environment, which in turn emits a reward and a new state. If the

application qubit is teleported, the reward is set to teleportation fidelity and the episode is terminated. As in the reset function, the state is advanced until an agent decision is necessary.

4.3.2 *Agent*

Implementing the agent is done by using the PPO Module from Stable-Baselines 3. It works with the environment above by taking an action, observing the resulting state and reward and updating a policy with the experience gathered. This is done in combination with the reward only being given when an application qubit is teleported, the reward for all other steps is 0 and an episode ends after application qubit is teleported. The idea is that all steps of an episode are equally important to the result. We don't want steps taken closer to the reward to have more influence on the training than the ones taken earlier.

For the other parameters such as `n_steps`, `learning_rate`, `batch_size`, `n_epochs` and `gae_lambda` a parameter optimization is done using the framework optuna [2].

The policy and the value function are given by a neural network with 2 hidden layers each having 256 nodes and tanh activation function.

4.3.3 *Agent Implementation and Hyperparameters*

The reinforcement learning agent is implemented using the PPO module from the Stable-Baselines 3 library [29]. The agent operates within the defined environment by executing actions, observing the resulting state transitions and rewards, and subsequently updating its policy based on the gathered experience. To align the training objective with the system's utility, a sparse reward structure is utilized where a non-zero reward is granted exclusively during the terminal step of an episode when an application qubit is successfully teleported. This reward value is set to the final teleportation fidelity. For all intermediate transitions, the reward is strictly zero, and the episode terminates immediately after the teleportation event.

The underlying philosophy for this sparse reward design is to ensure that every step within an episode is treated as equally significant to the final outcome. By avoiding incremental rewards, the framework prevents a bias where actions taken chronologically closer to the teleportation event exert a disproportionate influence on the policy update compared to earlier, foundational decisions. For the neural network architecture, both the policy and the value function are modeled using two hidden layers, each consisting of 256 nodes and employing a tanh activation function.

The remaining hyperparameter, specifically `n_steps`, `learning_rate`, `batch_size`, `n_epochs`, and `gae_lambda`, are determined through a systematic optimization process using the Optuna framework [2]. This framework navigates a defined state parameter space to identify the configuration that maximizes the agent's performance.

4.4 LEARNING PROCESS

The learning process is carried out by letting the agent interact with the Environment and evaluating the policy every 20000 steps. One evaluation takes 70 episodes. If the model is better than the one saved before it is replaced by the new one. If over 200 consecutive

evaluations the training did not result in a new best model, the training is stopped. To avoid stopping training too early there is a warm up period where training is not stopped for 100 episodes.

EVALUATION

For all evaluations we assume a cable length l of 5km between Node A and B. Thus the time between entanglement arrivals is

$$\Delta = \frac{l}{c} = 0.1ms \quad (5.1)$$

We assume that the cable has an attenuation of -0.15 db/km [12]. In combination with the cable length, the probability of failed generation is the probability of photon absorption in the optical fiber. Thus the success probability is given by

$$1 - p_g = 1 - e^{-\eta l} = 0.3047 \quad (5.2)$$

The ratio of the mean application qubit arrival time to the mean entanglement generation time to 100. For 1 application qubit arriving, 100 entanglements are generated. The queue length is set to 1.

5.1 BASELINE

First this work will begin with understanding and evaluating the purification protocols from Table 4.1. This will not include any training. Its purpose is to get baseline values that can be used for comparisons and creating an understanding of how the protocols work with different error types, defined by the values of the lambdas of the given Entanglement E . The error types will be evaluated for different coherence times t_c . For larger coherence times the degradation of the qubits and entanglements is slower. They stay in their generated state for longer. This becomes evident when recalling that the decay of a single qubit fidelity is described $\bar{F}_q(t) = (e^{-\frac{t-t_0}{t_c}} + 2)/3$ which shows a inverse correlation with t_c . Also the entanglement fidelity decay described by $F_l(t) = e^{-\frac{t-t_0}{t_c}} (F_l(t_0) - 0.25) + 0.25$ shows the same correlation.

5.1.1 Entanglements With Single Error Type

To make the analysis easier and see how the different protocols work with different error types, we start with Entanglements $E(\lambda_1, \lambda_2, \lambda_3)$ where we set 2 of the lambdas to 0. The remaining $\lambda_x > 0$ is called the active lambda λ_{active} . Table 5.1 shows the simplified formulas for the bilocal Clifford Protocols. For all three we can say that for PROT_x if λ_x is active the success probability only depends on fidelity of existing entanglement. This error type does not influence this purification protocol. A high error rate in this lambda does not lower the success probability of using the purification protocol. Regarding the jump function, a higher fidelity of the existing entanglement results in a higher fidelity. We call this configuration the matched case. For example for PROT_1 setting $\lambda_{2,3}$ to 0 is the matched case, as $\lambda_1 > 0$.

Protocol	active λ	Jump Function	Success Probability
PROT_1	λ_1	$\frac{(4\lambda_1-3)F_l-\lambda_1}{-2F_l-1}$	$\frac{2}{3}F_l + \frac{1}{3}$
	λ_2	$\frac{3(\lambda_2-1)F_l}{(4\lambda_2-2)F_l-\lambda_2-1}$	$\frac{2}{3}(1-2\lambda_2)F_l + \frac{1}{3}(1+\lambda_2)$
	λ_3	$\frac{3(\lambda_3-1)F_l}{(4\lambda_3-2)F_l-\lambda_3-1}$	$\frac{2}{3}(1-2\lambda_3)F_l + \frac{1}{3}(1+\lambda_3)$
PROT_2	λ_1	$\frac{3(\lambda_1-1)F_l}{(4\lambda_1-2)F_l-\lambda_1-1}$	$\frac{2}{3}(1-2\lambda_1)F_l + \frac{1}{3}(1+\lambda_1)$
	λ_2	$\frac{(4\lambda_2-3)F_l-\lambda_2}{-2F_l-1}$	$\frac{2}{3}F_l + \frac{1}{3}$
	λ_3	$\frac{3(\lambda_3-1)F_l}{(4\lambda_3-2)F_l-\lambda_3-1}$	$\frac{2}{3}(1-2\lambda_3)F_l + \frac{1}{3}(1+\lambda_3)$
PROT_3	λ_1	$\frac{3(\lambda_1-1)F_l}{(4\lambda_1-2)F_l-\lambda_1-1}$	$\frac{2}{3}(1-2\lambda_1)F_l + \frac{1}{3}(1+\lambda_1)$
	λ_2	$\frac{3(\lambda_2-1)F_l}{(4\lambda_2-2)F_l-\lambda_2-1}$	$\frac{2}{3}(1-2\lambda_2)F_l + \frac{1}{3}(1+\lambda_2)$
	λ_3	$\frac{(4\lambda_3-3)F_l-\lambda_3}{-2F_l-1}$	$\frac{2}{3}F_l + \frac{1}{3}$

Table 5.1: Simplified jump functions and success probabilities for PROT_1-3

Matched lambda is denoted λ_{active}

Otherwise, for PROT_x if λ_x is not the active lambda, the success probability depends on λ_{active} . A high active lambda lowers the success probability of purification as in $\frac{2}{3}(1-2\lambda_3)F_l + \frac{1}{3}(1+\lambda_3)$ the 2λ has more influence than λ . This is the case for both inactive functions. We can treat them as one, as they become the same.

To see how the success probability and jump function influence the average teleportation fidelity an evaluation using the simulation is carried out. The results are shown in [Figure 5.1](#) (c). The matched case has a worse average teleportation fidelity for all evaluated coherence times. It seems that the higher success probability of the matched case does not make up for the lower jump function result. So for example if we have an environment with Entanglements $E(0.3, 0.0, 0.0)$, then always choosing PROT_1 yields the worst performance. Conversely, if we always choose each PROT_2 or PROT_3 we get the same result twice, that is better than always choosing PROT_1. PMD has the highest resulting fidelity, but also the lowest success probability. In terms of avg. teleportation fidelity it yields the best performance, averaging a higher teleportation fidelity then all other purification protocols. But with PMD, we need to keep in mind, that it we can only use it when λ_{2-3} are set to 0. Thus it has the best performance for the error type it works on, but is not that flexible as the bilocal Clifford protocols.

5.1.2 Entanglements With Multiple Error Types

To understand the protocols more, we turn up the complexity. We now evaluate for entanglements where the lambda values are permutations of $(0.2, 0.1, 0.0)$. PMD can not be considered in this case. The bilocal Clifford protocols are highly symmetric. There are three cases to consider that each are true for all three protocols. We can assign each 0.2, 0.1, 0.0

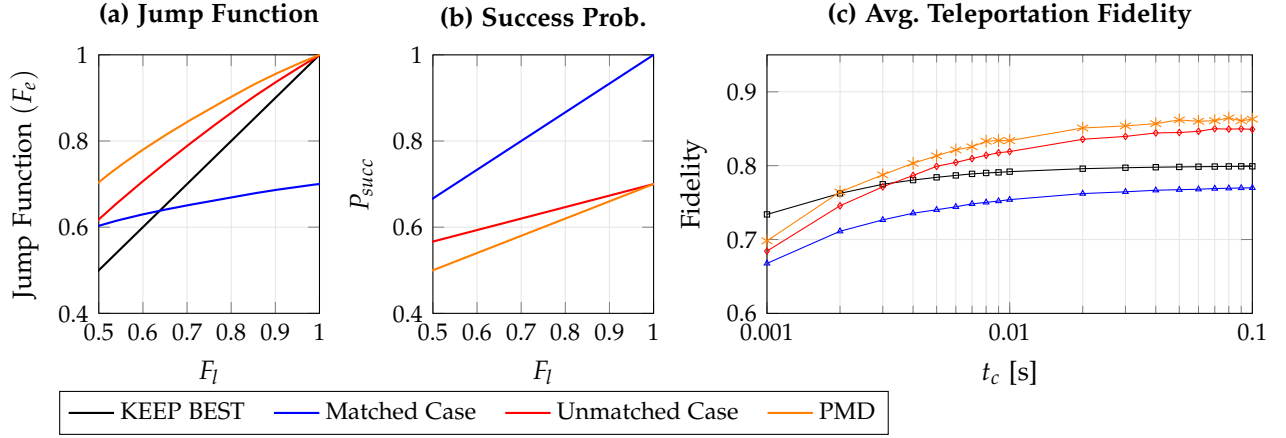


Figure 5.1: Jump function, success probability and avg. teleportation fidelity for entanglements E with $F(E) = 0.7$ and $\lambda_{active} = 0.3$.

to λ_x for PROT_x. Then we can assign the other two values to the lambdas left. Because these are both the unmatched case and are equal, the order has no effect on the result. Consequently there are only 3 permutations that yield different results. The two unmatched cases are now called Unmatched Case 1 and Unmatched Case 2.

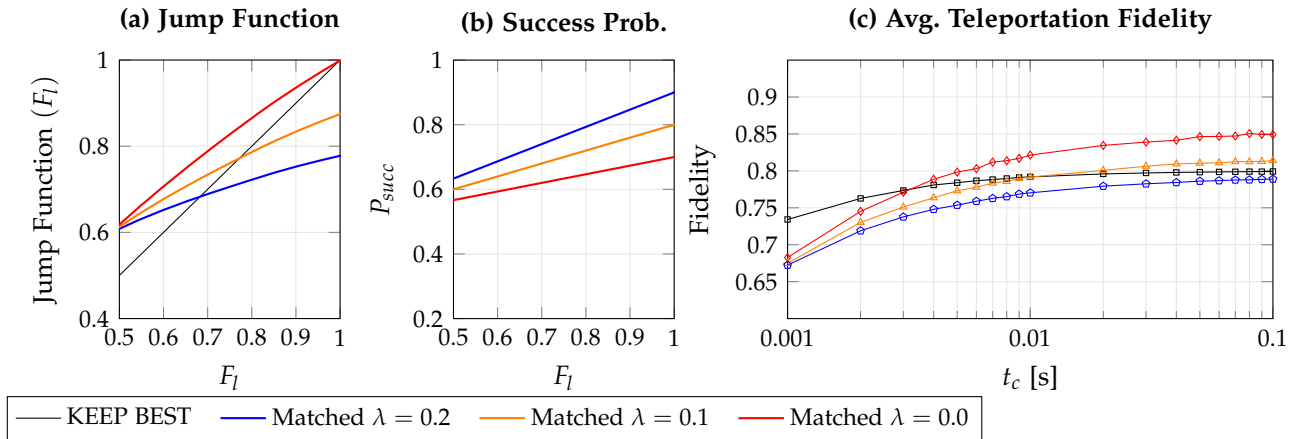


Figure 5.2: Comparison of the performance of $\lambda_{matched} = 0.2, 0.1$ or 0.0

The evaluation results are shown in Figure 5.2. $\lambda_{matched} = 0.2$ has the worst jump function, but best success probability, $\lambda_{matched} = 0.1$ has a moderate jump function and success probability and $\lambda_{matched} = 0.0$ has the best jump function and the worst success probability. We can see that $\lambda_{matched} = 0.0$ has the best average teleportation fidelity, second best is $\lambda_{matched} = 0.1$ and worst is $\lambda_{matched} = 0.2$. Again we see that the better jump function makes up for a worse success probability.

5.1.3 Influence Of Decoherence On Action Performance

When looking at [Figure 5.1\(c\)](#) and [Figure 5.2\(c\)](#) we clearly see that the coherence time as an influence on the average teleportation performance. Purification performs worse for smaller coherence times. When waiting for a new entanglement to perform purification the existing entanglements fidelity degrades fast. Once a new entanglement arrives, the existing entanglement might already be too bad for a good purification result. Inversely, for larger coherence rates, the fidelity of the entanglement in long-term memory stays good and can yield good performance when used for purification. KEEP BEST also performs better for larger coherence rates, as the noise becomes smaller. But in relation to the Purification protocols performance its performance becomes better for smaller coherence times. As shown, in [Figure 5.1\(c\)](#) and [Figure 5.2\(c\)](#), for a small enough coherence time, KEEP BEST outperforms the purification protocols. For larger coherence times, the purification protocols perform better.

We have an understanding of the purification protocols and KEEP BEST now. The results show that no single heuristic strategy performs optimally across all settings. In particular, strategies that always apply purification may achieve high fidelities in favorable error configurations but suffer from a higher risk of failure. Conversely, strategies that avoid purification achieve higher availability but result in lower average teleportation fidelity.

This highlights the need for adaptive strategies that dynamically balance these trade-offs depending on the current system state. We know want to evaluate the performance of the learned policies.

5.2 OPTIMAL POLICY

To find the optimal policy π^* we trained different models shown in [Table 5.3](#), with hyperparameter shown in [Table 5.2](#). The hyperparameter where found using a parameter optimization with parameter space

- **n_steps**: Categorical choice from the set $\{256, 512, 1024, 2048, 4096\}$
- **batch_size**: Categorical choice from $\{32, 64, 128, 256, 512, 1024\}$, subsequently constrained to the largest divisor of $n_steps \times \text{environments}$ used for training
- **n_epochs**: Integer range from 3 to 15
- **learning_rate**: Logarithmically scaled float range from 1×10^{-5} to 1×10^{-3}
- **gae_lambda**: Float range from 0.9 to 1.0
- **ent_coef**: Logarithmically scaled float range from 1×10^{-8} to 0.1.

on a specific environment setting. For parameter set 1, optuna was given the 4-GPS actions and entanglements $E(0.3, 0.0, 0.0)$ and for set 2, the actions from 5-GPS and the same entanglements.

A model entry in the table means that this training setting was used to train a dedicated model for all coherence times $t_c \in 0.001, 0.002, \dots, 0.01$ in steps of 0.001, as well as for $\{0.02, 0.03, \dots, 0.1\}$ in steps of 0.01. Generally, all models could use the KEEP BEST action, as it is an indispensable operation to control the fidelity-availability trade-off. Also, all

models worked on entanglements E with initial fidelity $F_e = 0.7$. The 5-GPS model could use all possible actions and thus the expectation was that it yields the best performance. Especially the comparison to the 2-GPS (PMD) model, using the method proposed in [24] is crucial, because this is the test whether having more protocols available results in a better policy π . As PMD can only be used for Entanglements with $E(\lambda_1, 0, 0)$ the models 4-GPS where trained using only KEEP BEST and the bilocal Clifford protocols as these are more flexible in terms of error types they work on. 3 models where trained, each on one of the Error types $E(0.3, 0.0, 0.0)$, $E(0.0, 0.3, 0.0)$, $E(0.0, 0.0, 0.3)$. But as shown in Figure 5.1, PMD outperforms the protocols in terms of average teleportation fidelity, so we do not expect 4-GPS to outperform 5-GPS in that domain. Again, to test our assumptions of using more protocols to find a better policy for controlling the fidelity-availability trade-off, we trained the model 2-GPS (PROT_2) compared it to the 4-GPS model trained on $E(0.3, 0.0, 0.0)$. Additionally, we trained 4-GPS (RANDOM), that could access all bilocal Clifford protocols and received Entanglements with fidelity 0.7 and random distribution of the remaining 0.3 error probability on λ_{1-3} . The goal is to have a single model performing as good as the three 4-GPS models, without having to train each separately. If the performance is as good it would yield an efficiency gain, as less models are needed to achieve the same results.

Hyperparameter	Set 1	Set 2
n_steps	1024	512
batch_size	256	128
n_epochs	6	15
learning_rate	2.9496×10^{-4}	4.622×10^{-4}
γ (gamma)	1	1
gae_lambda	0.99	0.921
ent_coef	0.0030118	1.874×10^{-7}

Table 5.2: PPO Hyperparameter

Model	Actions	HP Set	Environment
5-GPS	PMD, PROT_1, PROT_2, PROT_3, KEEP BEST	2	$E(0.3, 0.0, 0.0)$
3 x 4-GPS	PROT_1, PROT_2, PROT_3, KEEP BEST	1	$E(0.3, 0.0, 0.0)$, $E(0.0, 0.3, 0.0)$, $E(0.0, 0.0, 0.3)$
4-GPS (RANDOM)	PROT_1, PROT_2, PROT_3, KEEP BEST	1	Random Errors ($\sum = 0.3$)
2-GPS (PMD)	PMD, KEEP BEST	1/2	$E(0.3, 0.0, 0.0)$
2-GPS (PROT_2)	PROT_2, KEEP BEST	1	$E(0.3, 0.0, 0.0)$

Table 5.3: Model Configurations

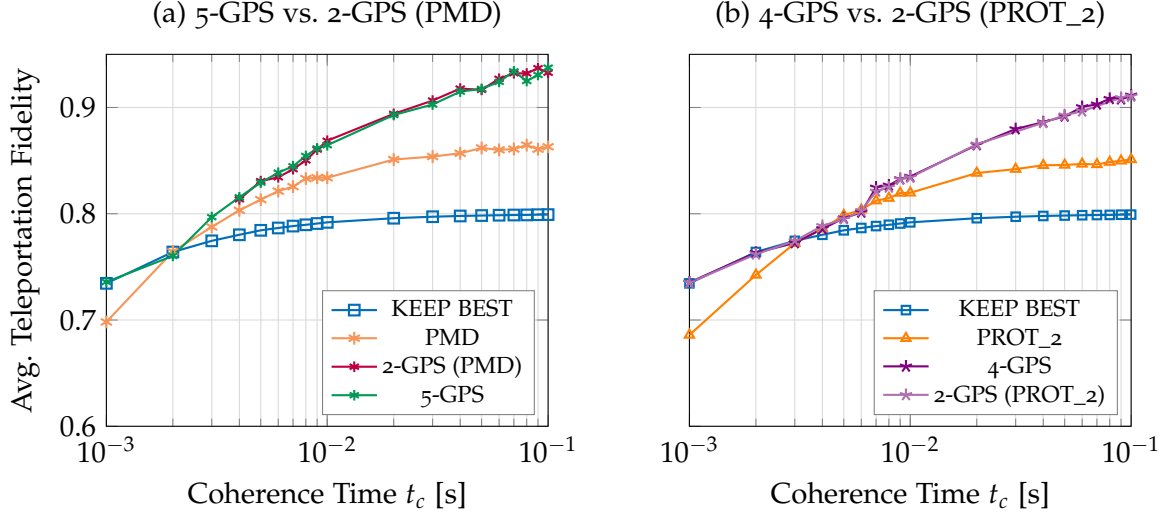


Figure 5.3: (a) shows 5-GPS vs 2-GPS (PMD) performance. (b) shows 4-GPS vs 2-GPS (PROT_2) performance. Both include baseline strategies.

5.2.1 Evaluating For Different Coherence Times And Error Types

We begin with evaluating for the utility $U = \mathbb{E}[\bar{F}_T(F_l, \bar{F}_q(t, \kappa))]$ assuming a waiting time sensitivity of $\kappa = 1$. We focus on how the approach of integrating multiple purification protocols performs across various coherence times and error configurations.

Figure 5.3(a) compares the 5-GPS model to the heuristic baselines KEEP BEST and PMD, as well as the 2-GPS (PMD) model. Comparing the adaptive strategies, neither strictly outperforms the other across all coherence times; a universally superior policy does not emerge. For low coherence times ($t_c < 0.003$), both 5-GPS and 2-GPS (PMD) perform as well as the KEEP BEST strategy. In this regime, decoherence dominates the system behavior, causing entanglements to degrade rapidly. As a result, purification becomes ineffective, since the fidelity loss during waiting outweighs the potential gains from successful purification. Consequently, strategies that avoid purification and prioritize availability, such as KEEP BEST, are sufficient.

However, for higher coherence times, the adaptive models surpass both PMD and KEEP BEST. As the coherence time increases, entanglements remain stable for longer periods, allowing the agent to apply purification more effectively. In this regime, the learned policies are able to exploit the additional time to improve fidelity while still maintaining acceptable availability. As a result, the performance gap between the static PMD baseline and the adaptive strategies widens, demonstrating that the learned policy yields greater benefits in more stable environments.

Nevertheless, it becomes evident that incorporating additional purification protocols (5-GPS) does not provide a noticeable advantage in average teleportation fidelity over the 2-GPS (PMD) approach, which represents the strategy proposed in [24]. For Figure 5.3(b), we observe a similar result. Although both policies 4-GPS and 2-GPS (PROT_2) outperform the baseline strategies PROT_2 and KEEP BEST for $t_c > 0.006$, there is no benefit in using more protocols, as 4-GPS does not perform better than 2-GPS (PROT_2). Additionally, we observe that the coherence time threshold at which the adaptive strategies begin to outperform the

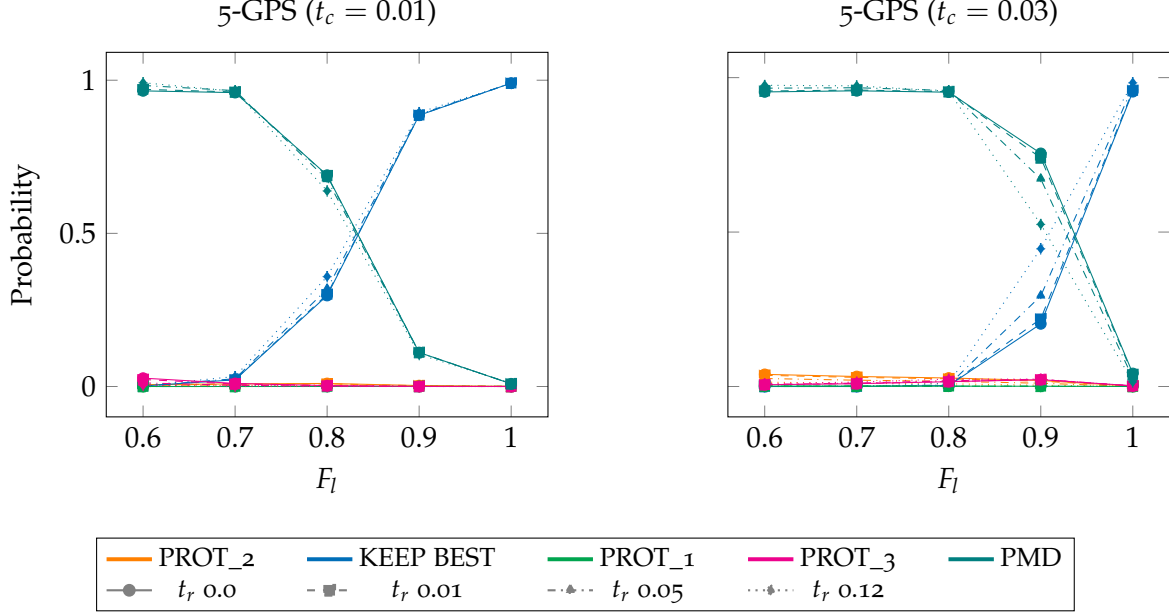


Figure 5.4: Action probabilities for 5-GPS for coherence times 0.01 and 0.03).

baseline strategies in (a) is higher compared to (b). This can be attributed to the fact that PMD performs better than PROT_2 in the considered error configuration.

Overall, these results highlight two important insights. First, the effectiveness of adaptive policies strongly depends on the coherence time, as it determines whether purification is beneficial at all. Second, increasing the number of available purification protocols does not necessarily improve performance in environments with fixed error types. Instead, the main advantage of adaptive strategies lies in their ability to dynamically balance the trade-off between fidelity and availability, rather than in the size of the action space.

5.2.1.1 General Strategy

But how do the models generally achieve a performance gain in comparison to strategies always applying the same action? To find out we compare the action probabilities of the models for different coherence times, beginning with $t_c = 0.01$ in Figure 5.4. When comparing the action probabilities for $t_r = 0$ and changing F_l we see that for low values $F_l = 0.6$ or $F_l = 0.7$, exclusively PMD is used. This is due to the model trying to increase F_l through purification, because the current fidelity is not optimal. Additionally, because the fidelity is not optimal, there is less of a risk of losing the entanglement, because a newly generated one has a comparable fidelity. Thus in terms of controlling the fidelity-availability trade-off to optimize U , it tries to optimize for fidelity. For higher values for F_l , the probability shifts more towards using KEEP BEST. This can be accounted to the model now having an entanglement with significantly better fidelity than what newly generated have. Losing it would mean having to apply purification for a new entanglement again. Moreover, there is less fidelity to gain, as the maximum fidelity achievable is 1. Consequently, for $F_l = 1$, the model exclusively applies KEEP BEST. Thus the policy optimizes the trade-off by focusing on the availability. The time since last request t_r has only a small influence on action probabilities, making the model lean more towards KEEP BEST. The influence of

t_r can be better observed for higher coherence values. For example for $t_c = 0.03$, the wait time has a significant impact on the probability distribution between KEEP BEST and PMD. This is because changing the t_r , changes how the policy approaches the fidelity-availability trade-off. The higher t_r the more the policy optimizes towards availability through a higher likelihood of applying KEEP BEST. The model avoids purification, because if this fails, there is no time left to improve the fidelity again.

When comparing the 5-GPS model at $t_c = 0.01$ to the one at $t_c = 0.03$, we observe a shift in fidelity where the KEEP BEST strategy becomes favored over PMD. This transition occurs because entanglement degrades more slowly at higher coherence times, making it feasible to achieve and maintain higher entanglement fidelities. Consequently, the model learns that it can realistically target these higher fidelities with reduced risk, ultimately improving the average teleportation fidelity. Furthermore, the loss of a high-fidelity entangled pair becomes less detrimental, as the extended coherence time facilitates the regeneration of such states. We furtherance the strategy of the 5-GPS model by observing the relationship between the coherence time and the threshold upon which KEEP BEST is used, which is shown in Figure 5.5.

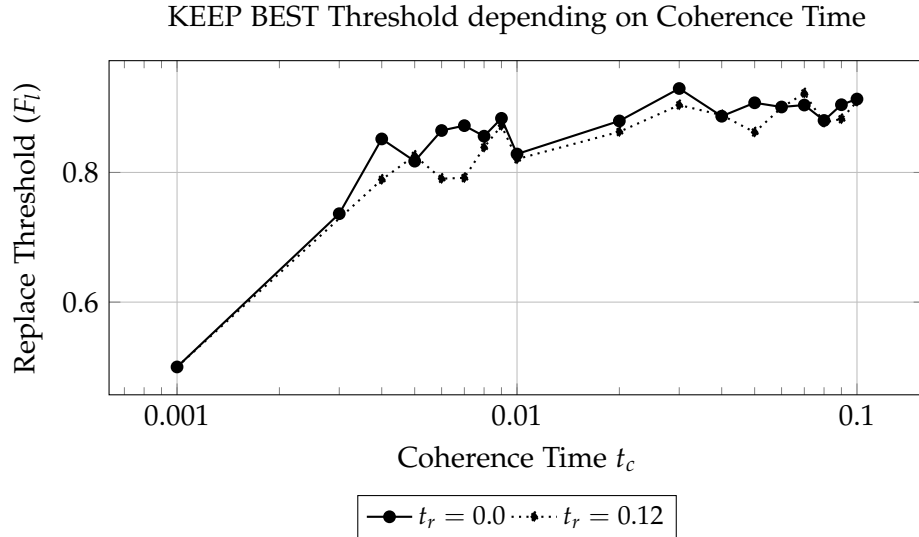


Figure 5.5: F_l threshold upon which KEEP BEST is used for 5-GPS for different coherence times t_c and time since last request t_r values

We see that although there is a trend for increasing threshold from smallest to largest coherence time, it is not increasing monotonically. The graph offers more insight into how the strategy of 5-GPS changes for coherence times. For $t_c = 0.001$, the model only uses KEEP BEST, as it uses KEEP BEST for all entanglements with fidelity larger or equal to 0.5 and below that purification is not possible. This shows that purification provides no value for lower coherence times and always applying KEEP BEST is the best strategy. Consequently the model achieves the same fidelity as KEEP BEST. For $t_c = 0.002$ it never applies KEEP BEST. For coherence values larger than that, the model always has a threshold upon which KEEP BEST is applied. A higher time since last request value of $t_r = 0.12$ decreases the threshold for all values but $t_c = 0.007$. Apparently, for $t_c = 0.007$, the fidelity than can be gained with purification outweighs the risk of having no entanglement available upon

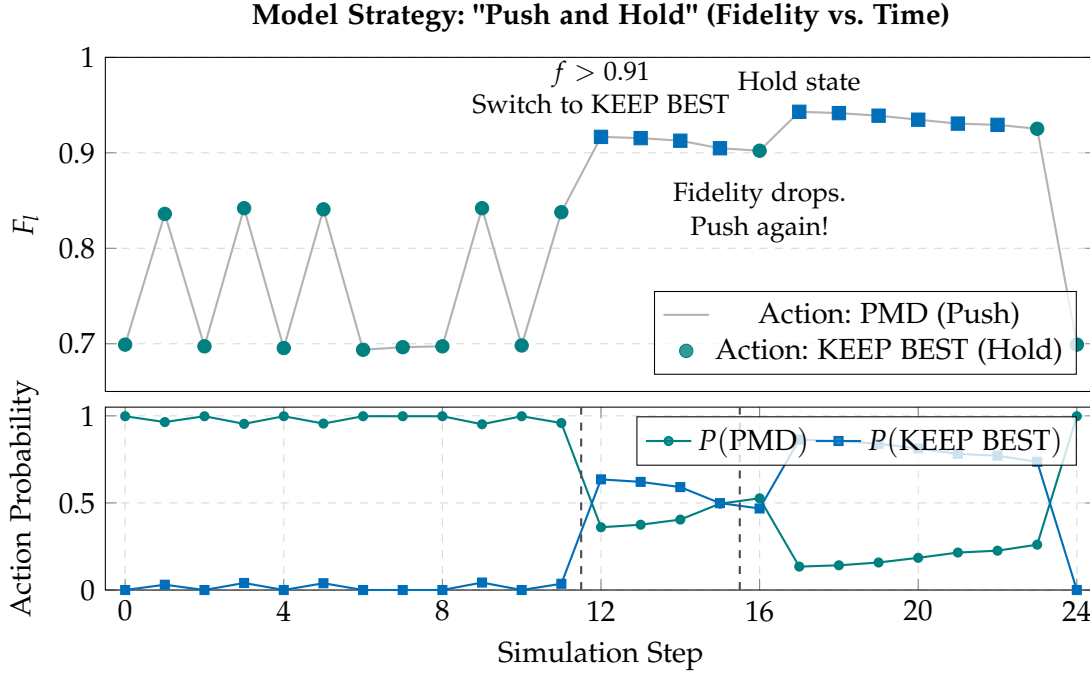


Figure 5.6: Demonstration of the "Push and Hold" strategy applied by the 5-GPS model in a simulation

application qubit arrival. This is an interesting outlier and could hint to an alternative strategy, that works different to the way all the other coherence times are handled.

The observed strategy can be described by "Push and Hold". An example of the 5-GPS model applying the actions in a simulation is shown in Figure 5.6. We see that for the first 11 steps PMD is applied. In the first 7 steps we see an alternating sequence of successes and failures, starting with a success. We need to recall, that when long-term memory is empty, then a newly generated entanglement is instantly placed there, without the agent needing to make a decision. That is why after the second step, which is a failed purification attempt, the model takes the next action, once a new entanglement is already placed in long-term memory again. Only after two consecutive successful purification, we see the model switching to KEEP BEST to keep the high fidelity entanglement. Following that we see decoherence degrading the fidelity up to the point, where the model applies purification again, to achieve a high fidelity again.

5.2.1.2 Impact Of Extended Action Spaces On Policy Behavior

A central question of this evaluation is whether providing the agent with a larger set of purification protocols (as in the 5-GPS and 4-GPS models) yields a higher average teleportation fidelity U compared to an agent restricted to a single purification protocol alongside the KEEP BEST strategy (2-GPS).

As observed in Figure 5.3, models with extended action spaces do not outperform their 2-GPS counterparts in environments with a static error type. While the additional protocols increase the theoretical flexibility of the agent, this does not translate into a measurable performance improvement. To understand this behavior, we analyze the learned action

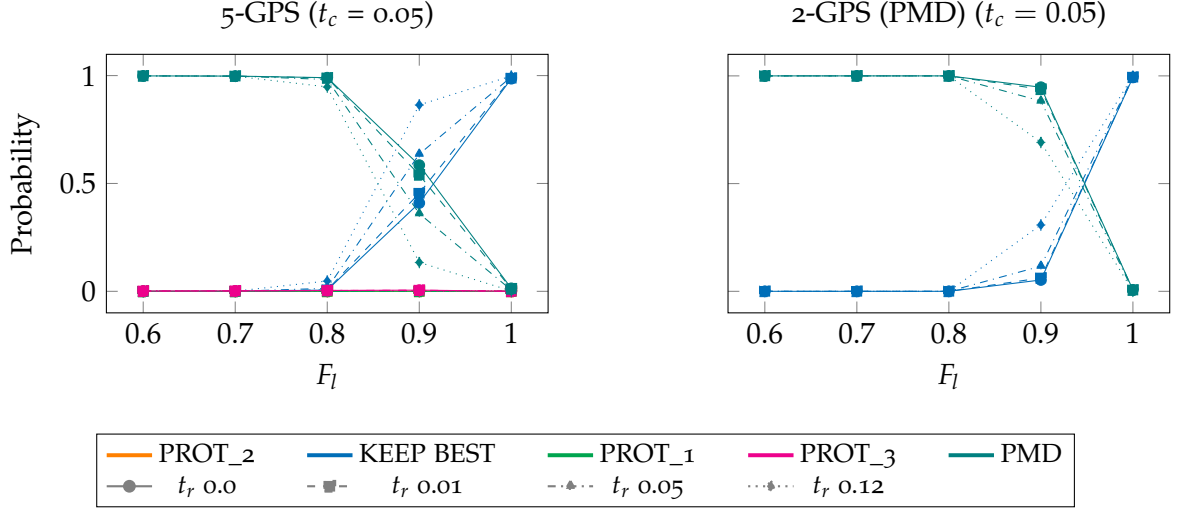


Figure 5.7: Action probability values of the models 5-GPS and 2-GPS (PMD) for $t_c = 0.05$ and different values of F_l and t_r

probabilities for $t_c = 0.05$ across varying long-term memory fidelities F_l and request times t_r .

Comparing the action distributions of the 5-GPS and 2-GPS (PMD) models in Figure 5.7, we observe that the 5-GPS policy effectively converges to a 2-GPS strategy, exclusively utilizing PMD and KEEP BEST. The probability of selecting PROT_1, PROT_2, or PROT_3 drops close to 0. During training, the agent learns that for the fixed error model $E(0.3, 0.0, 0.0)$, PMD provides the best trade-off between fidelity improvement and success probability. As a result, the reinforcement learning algorithm performs an implicit form of action pruning, discarding the suboptimal Clifford protocols in order to maximize the expected utility U .

For 4-GPS and 2-GPS (PROT_2), we observe a similar result in Figure 5.8. Both models learn the same “Push and Hold” strategy, alternating between purification and preservation depending on the system state. Looking at the 4-GPS policy, we again observe a close to 0 probability of selecting PROT_1, indicating that this action does not contribute to optimizing U in the given environment. This is consistent with the dominance of the λ_1 error component in $E(0.3, 0.0, 0.0)$, which makes the matched protocol ineffective in terms of resulting fidelity improvement. The high success probability of PROT_1 does not seem to provide a benefit in the found strategy. Both 5-GPS and 4-GPS rely on the purification protocol with the highest resulting fidelity and lowest success probability in favor of other purification protocols with a different trade-off.

For PROT_2 and PROT_3, we know that for entanglements $E(0.3, 0.0, 0.0)$ they exhibit identical behavior, as shown in Table 5.1. Consequently, it is expected that the learned policy assigns similar probabilities to both actions, as they are equally suitable for improving fidelity. This further explains why the extended action space does not provide an advantage: even though multiple actions are available, two of them are effectively redundant in this setting.

Overall, these results show that in environments with fixed error types, the optimal policy can be expressed using only a subset of the available actions. The additional protocols in the extended action space do not improve performance, as the agent converges to a reduced

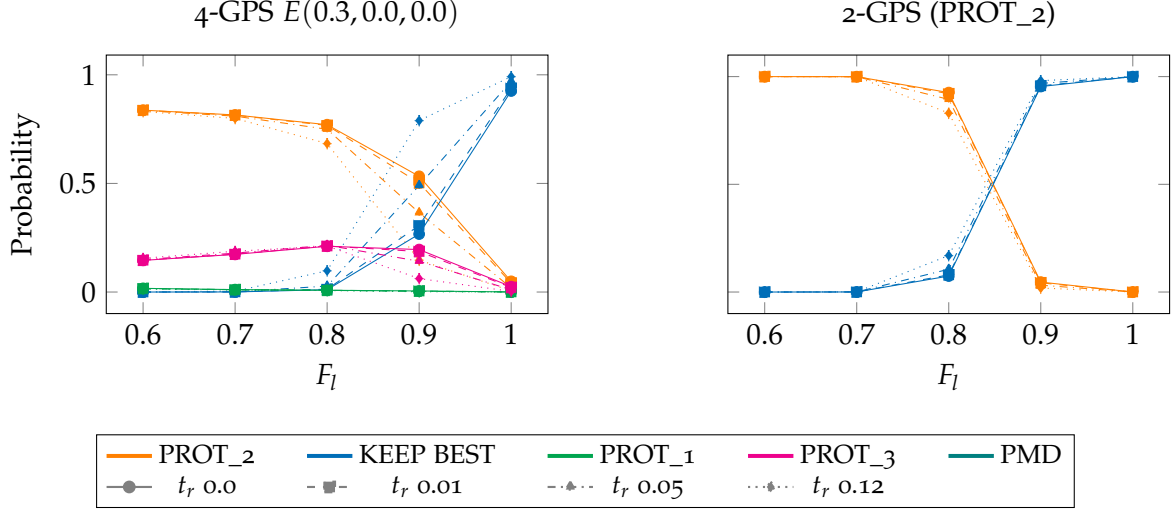


Figure 5.8: Action probability values for $E(0.3,0.0,0.0)$ and 2-GPS (PROT_2) at $t_c = 0.05$ across different values of F_l and t_r

policy that only includes the most effective actions. This highlights that the benefit of larger action spaces is highly dependent on the complexity of the environment and may only become relevant in more dynamic settings, such as those with varying error distributions.

5.2.2 4-GPS For Different Error Types

In addition to training the model 4-GPS $E(0.3,0.0,0.0)$, training the 4-GPS $E(0.0,0.3,0.0)$ and 4-GPS $E(0.0,0.0,0.3)$ models also worked as shown in Figure 5.9. We observe that the two additional models can achieve a similar result for U . This indicates that the fidelity-availability trade-off can be controlled across multiple entanglement types, resulting in a wider range of scenarios in which the approach can be effectively applied. In particular, it shows that the proposed reinforcement learning framework is not limited to a single dominant error configuration but can adapt to different underlying noise structures.

Strategy-wise, we see in Figure 5.10 that all three models learned to not use the matched protocol. For example, 4-GPS $E(0.3,0.0,0.0)$ does not use PROT_1, which corresponds to the dominant error component. This behavior is consistent across all evaluated error types and aligns with the theoretical understanding of purification protocols, where matched protocols tend to provide less favorable improvements under dominant error conditions. Instead, the models rely on the remaining protocols, which yield better trade-offs between fidelity increase and success probability in these settings.

Additionally, all models use KEEP BEST in a similar way, following the “Push and Hold” strategy. In this strategy, purification is applied to increase the fidelity of the entanglement (push phase), followed by a phase where the current entanglement is preserved using KEEP BEST (hold phase). While the overall structure of this strategy is consistent, the exact probabilities differ slightly between models, especially for higher values of F_l , such as $F_l = 0.9$. However, these differences do not significantly impact the resulting average teleportation fidelity.

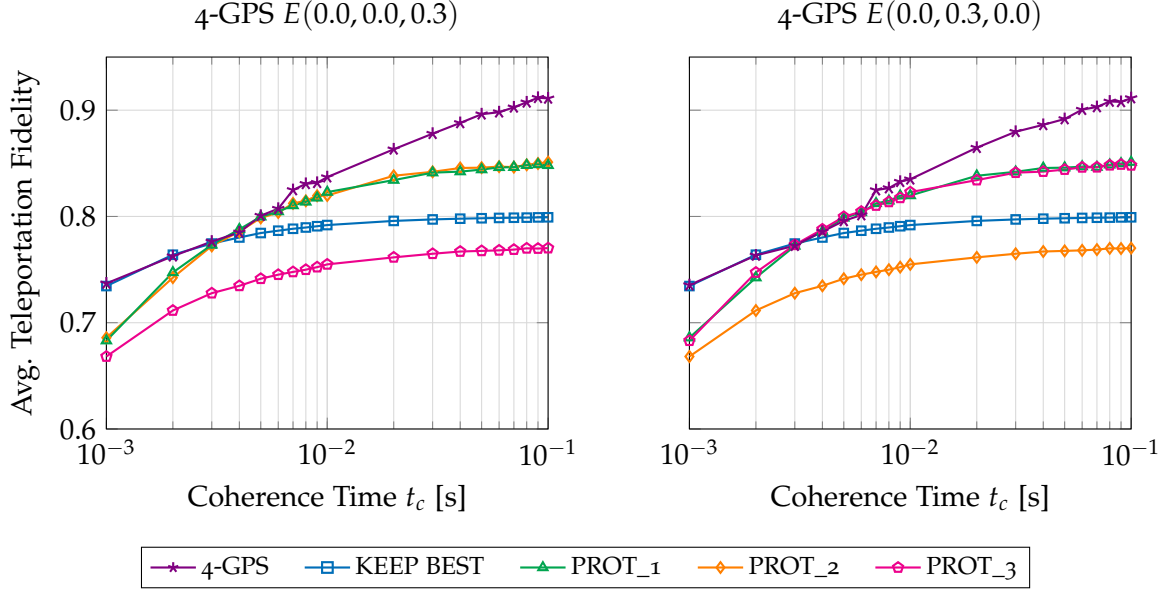


Figure 5.9: Comparison of 4-GPS performance for alternative error models. The baseline strategies are fully depicted and remain consistent across all evaluated scenarios.

Interestingly, the results suggest that the exact threshold at which KEEP BEST is applied does not strongly influence the average teleportation fidelity, as all three 4-GPS models achieve similar performance. This indicates that there exists a range of near-optimal policies that differ in their precise switching behavior between purification and preservation, but still result in comparable outcomes.

For 4-GPS $E(0.3,0.0,0.0)$, we have shown that the same average teleportation fidelity can also be achieved with fewer available protocols per entanglement type. Combining this observation with the results of the 2-GPS (PROT_2) model, and considering that the bilocal Clifford protocols are highly symmetric, we can generalize this finding. For $E(0.0,0.3,0.0)$, a model using only KEEP BEST and PROT_1/3 would be sufficient, and for $E(0.0,0.0,0.3)$, a model using only KEEP BEST and PROT_1/2 could achieve the same result.

This further supports the idea that the learning problem is structurally similar across different error types and that the agent is able to implicitly identify and utilize only the relevant subset of actions. Consequently, while the extended action space provides flexibility, it is not strictly required for optimal performance in environments with fixed error distributions.

5.2.2.1 4-GPS For Random Errors

We have shown that we can create policies for improving Avg. teleportation fidelity using purification for different fixed error types. Until now, we trained one model for each error type. In practice this requires lots of time. We want to improve efficiency by trying to train a single model that can work on all error types for entanglements with fidelity 0.7. To test this, we train models on entanglements with fidelity 0.7 and the remaining error of 0.3 distributed randomly on λ_{1-3} . The model has the ability to use all three bilocal Clifford protocols, as these can be applied on all on all error types. PMD is not included, as it does not have this ability. We keep both the best model acquired by saving the model having

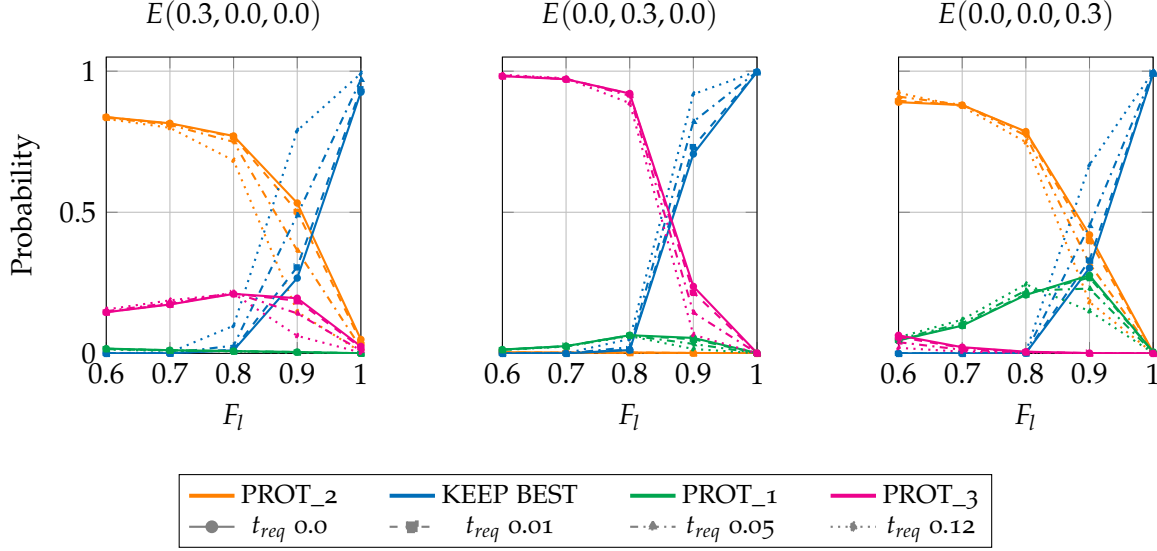


Figure 5.10: Action probabilities for the 4-GPS models, depending on F_l and time since last request for entanglements with different errors and $t_c = 0.05$.

the best reward in training and the end model, which might be worse for the environment the model was tested on. But it might be better for the other environments, because it will likely see more states, as training keeps going on for lots of episodes before it ends. In the end, we keep the model yielding the better results and name it 4-GPS (Random).

When looking at Figure 5.11 we see that generally, the performance of the in the environment with random errors is worse than in environments with fixed error, excluding the performance of the KEEP BEST strategy. KEEP BEST shows the same performance across all environments, which is as expected, as it is not dependent on the type of error. Only changing the fidelity of generated entanglements would change the performance of KEEP BEST.

Moreover, we see that for Random errors, PROT_1-3 perform equally good. This highlights, that the different bilocal Clifford protocols each only give an advantage over the others, if the error type lends in one direction. If over time, all errors occur equally often, then all protocols perform equally good. In comparison to KEEP BEST the protocols perform worse than in the case with a single error type, showing that the protocols on their own yield less improvement in comparison to KEEP BEST.

4-GPS (Random) performs as good as KEEP BEST for $t_c < 0.005$. For higher coherence values, we see the model outperforming the baseline strategies. and an increasing average teleportation fidelity with higher coherence values. In comparison to the environments with fixed errors, the model generally performed worse, which can be attributed to the lesser effect the purification protocols have. At every purification step the model applies a

From a reinforcement learning perspective, this setting introduces a significantly more complex learning problem. The agent is no longer able to rely on a fixed mapping between states and optimal actions, but must instead learn a policy that generalizes across a distribution of environments. This increases both the variance of the reward signal and the difficulty of credit assignment.

As a result, the learned policy represents a compromise between multiple optimal strategies, rather than specializing in a single one. This trade-off between specialization and generalization explains the observed performance gap compared to the models trained on fixed error types.

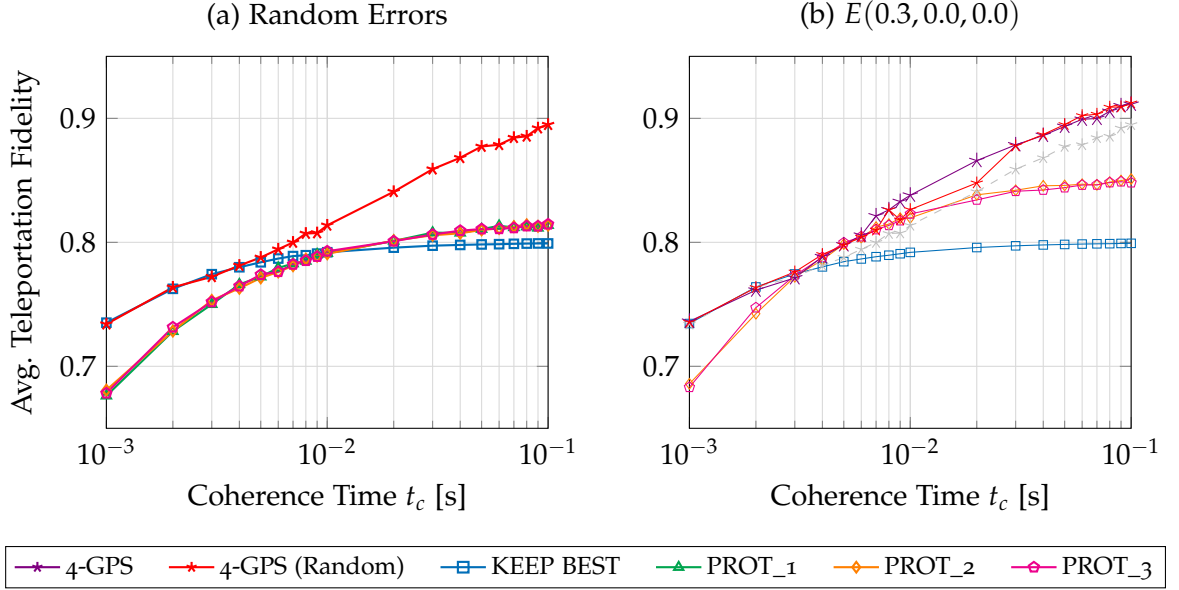


Figure 5.11: Comparison of 4-GPS performance for alternative error models.

We have seen that generally, the training worked for the environment with random errors. We now want to see how the model performs with dedicated errors. For coherence values larger than 0.02, we see that 4-GPS (Random) model performed as good as the models trained dedicated for one error type, as can be seen in Figure 5.11 (b) and Figure 5.12. For coherence values smaller than 0.007 4-GPS (Random) also yields the same performance. But for the coherence values $t_c = 0.007$ to $t_c = 0.02$, we do not see the model performing as good as the dedicated models consistently. Although there are cases where the model could perform good e.g. for $E(0.0, 0.3, 0.0)$ and $t_c = 0.01$ in most cases it only performed as good as the Purification strategy baselines or even worse as good as the KEEP BEST strategy. This is probably a problem of training the model correctly, not a problem of the environment making it impossible to learn, because the model has the same actions available as the dedicated models accomplishing the tasks and is given the same environment when being evaluated. A problem could be, that the 4-GPS (Random) model is not evaluated for Entanglements $E(0.3, 0.0, 0.0)$, $E(0.0, 0.3, 0.0)$ and $E(0.0, 0.0, 0.3)$ and thus does not necessarily need to learn the best strategies for these cases. It is only evaluated for random errors which could include enough Entanglements of these types to learn the strategy, but it is not necessarily the case. Strategy-wise we analyze whether it differs between the delicately trained models and the Random model for $t_c = 0.05$ in Figure 5.13, which is a coherence time for which the 4-GPS (Random) model consistently performed as well as the delicately trained models. Generally, the Push and Hold strategy is also applied here, as can be seen by how the KEEP BEST strategy is used. It is applied with higher probability for higher values of F_I . Also the model learned to not used the matched protocol, as PROT_x is never used when λ_x is active. For the other two protocols, the distinction on which protocol to use is not that clear. Two of the three models used both, depending on the policy.

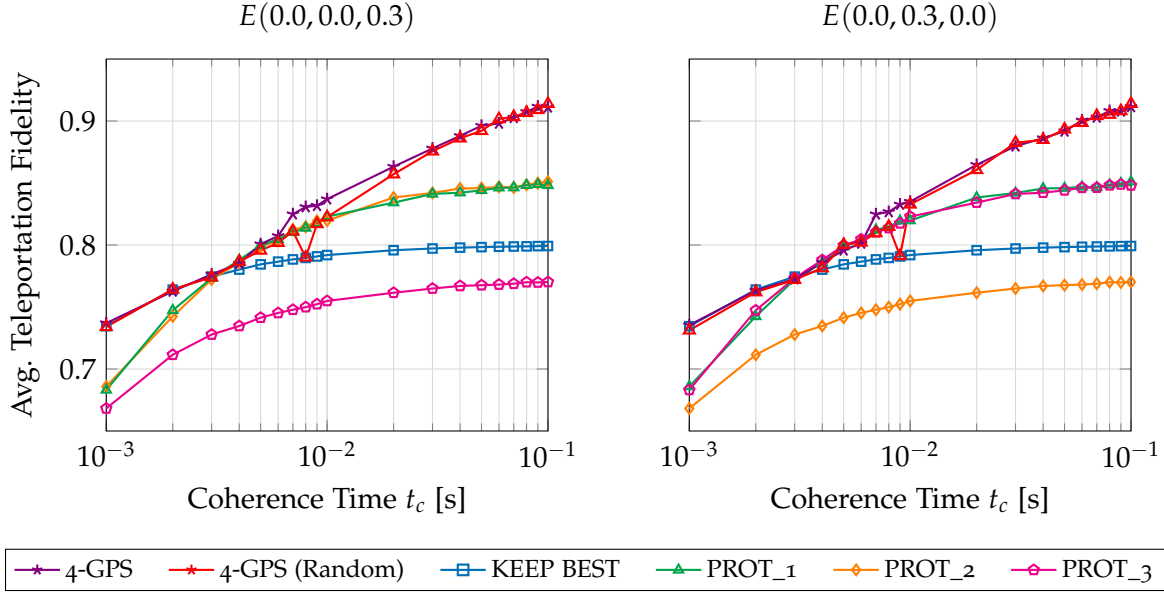


Figure 5.12: Comparison of 4-GPS performance for alternative error models. The baseline strategies are fully depicted and remain consistent across all evaluated scenarios.

To summarize, using a single model for different error types is possible for coherence values $t_c \in 0.001 - 0.005, 0.03 - 0.1$. For the other coherence times we did not find models that performed consistently well, but we also did not show that it is not possible.

DISCUSSION. The results reveal that the effectiveness of adaptive strategies is strongly dependent on the coherence time. In environments with low coherence times, the system dynamics limit the achievable performance, as entanglement degradation dominates and purification becomes ineffective. In such cases, simple heuristic strategies are already sufficient.

In contrast, for higher coherence times, the system allows for meaningful optimization, and adaptive policies are able to exploit the additional temporal stability. This demonstrates that the benefit of reinforcement learning is not uniform across all regimes, but rather emerges only when the environment provides sufficient opportunity for decision-making.

Furthermore, the results show that increasing the number of available purification protocols does not necessarily improve performance. Instead, the agent converges to a reduced set of effective actions, indicating that the structure of the environment plays a more important role than the size of the action space.

5.2.3 Evaluating For Different Wait Time Sensitivities

In addition to the influence of coherence time and error distributions, the sensitivity of application qubits to waiting time plays a crucial role in determining the optimal control strategy. This sensitivity is modeled by the parameter [Equation 4.4](#), which directly affects the rate at which the fidelity of application qubits degrades while waiting in the queue. As described earlier, larger values of κ correspond to applications that are highly sensitive to delay, whereas smaller values indicate more tolerant applications.

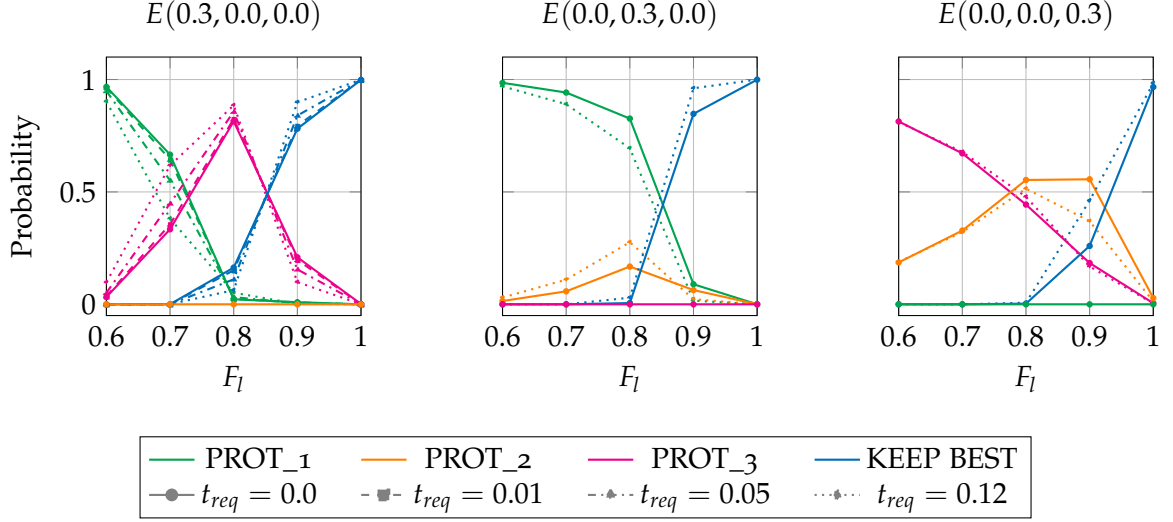


Figure 5.13: Action probabilities for 4-GPS (Random) for $t_c = 0.05$. The lines demonstrate how the agent shifts its policy from purification to replacement as the request ages (t_{req} increases), regardless of the error type (λ).

To evaluate the impact of different wait time sensitivities, we analyze the performance of the trained policies across a range of κ values while keeping all other system parameters fixed. This allows us to isolate the effect of waiting time sensitivity on the learned behavior and the resulting teleportation fidelity.

As shown in Figure 5.14, increasing κ leads to a noticeable decrease in the average teleportation fidelity across all evaluated models. This behavior is expected, as higher wait time sensitivity causes the fidelity of queued qubits to decay more rapidly, making delays significantly more costly.

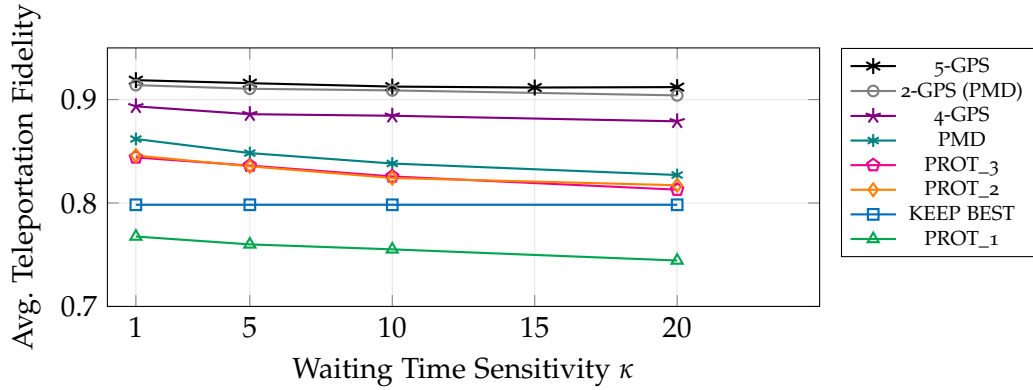


Figure 5.14: Comparison of teleportation fidelity across different waiting time sensitivities κ . The 5-GPS and 2-GPS models maintain superior stability compared to simpler protocols.

However, the adaptive policies (e.g., 5-GPS and 2-GPS models) consistently outperform static baseline strategies across all values of κ . This demonstrates that the learned policies are capable of adjusting their behavior to account for different application requirements. Instead,

the agent implicitly adapts through the reward signal, which reflects the degradation of qubit fidelity over time.

Another important result is that the adaptive models maintain relatively stable performance across varying κ values compared to baseline strategies. While all approaches experience a drop in fidelity as κ increases, the degradation is less pronounced for the RL-based policies. This indicates that the learned strategies generalize well across different application requirements and do not rely on a fixed trade-off assumption.

In contrast, static strategies such as always applying purification or always keeping the best entanglement cannot adapt to changing waiting time sensitivities. As a result, they either incur unnecessary delays (in the case of aggressive purification) or fail to achieve sufficient fidelity (in the case of conservative strategies).

CONCLUSION

In this work, we investigated the use of reinforcement learning for adaptive entanglement management in quantum communication systems. The goal was to dynamically balance the trade-off between teleportation fidelity and availability by selecting appropriate purification strategies under varying system conditions.

A central finding of this work is that increasing the number of available purification protocols does not necessarily lead to improved performance. In environments with fixed error types, the agent consistently learns to ignore suboptimal actions and converges to a reduced set of effective strategies. This indicates that the optimal policy is often dominated by a single purification protocol, making additional actions redundant. Consequently, the performance of the system is primarily limited by the environment dynamics rather than the size or expressiveness of the action space.

The evaluation demonstrates that reinforcement learning is capable of learning effective policies that adapt to the system state and outperform static baseline strategies in relevant regimes. In particular, the learned policies consistently exploit the trade-off between fidelity improvement and availability, resulting in higher average teleportation fidelities for sufficiently large coherence times.

A key insight is that the benefit of adaptive policies strongly depends on the structure of the environment. In regimes with low coherence times, where entanglement rapidly degrades, simple heuristic strategies such as KEEP BEST already achieve optimal performance. In contrast, for higher coherence times, the system provides sufficient stability for optimization, allowing reinforcement learning to significantly improve performance.

Across different fixed error types, the learned policies exhibit similar behavior due to the symmetry of the underlying purification protocols. This results in comparable performance across environments and suggests that the decision problem remains structurally consistent under permutation of the error parameters.

In more complex environments with random error distributions, the performance of the learned policies decreases compared to the fixed error case. This setting was specifically considered to enable the training of a single, generalized model that can be applied across different error configurations, instead of relying on multiple specialized models.+

While this approach proved successful for many parameter settings, the generalized model was not able to consistently match the performance of models trained on dedicated error types. This can be attributed to the increased difficulty of the learning problem. Due to the nonlinear structure of the bilocal Clifford purification protocols, the outcome of a given action depends strongly on the specific error configuration, leading to state-dependent and less predictable transitions. As a result, the agent is required to generalize across a wide range of heterogeneous environments, rather than optimizing for a single, well-defined setting. This significantly increases the complexity of the training process and. Consequently, the learned policy performs well across many scenarios, but does not achieve the same level of optimality as specialized policies in all cases.

Additionally, we demonstrated that the learned policies adapt to different waiting time sensitivities, effectively shifting the balance between fidelity optimization and availability depending on application requirements. This highlights the flexibility of the approach and its potential applicability to a wide range of use cases.

Overall, the results show that reinforcement learning provides a powerful framework for adaptive decision-making in quantum communication systems. However, we could not demonstrate that increasing the number of available purification protocols leads to a performance improvement. In the evaluated settings, the learned policies consistently converged to a reduced set of effective actions, indicating that additional protocols do not provide a measurable benefit in structured environments.

BIBLIOGRAPHY

- [1] Joshua Achiam. “Spinning Up in Deep Reinforcement Learning.” In: (2018).
- [2] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. “Optuna: A Next-generation Hyperparameter Optimization Framework.” In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2019.
- [3] Kai Arulkumaran, Marc Peter Deisenroth, Miles Brundage, and Anil Anthony Bharath. “Deep Reinforcement Learning: A Brief Survey.” In: *IEEE Signal Processing Magazine* 34.6 (Nov. 2017), pp. 26–38. ISSN: 1558-0792. DOI: [10.1109/MSP.2017.2743240](https://doi.org/10.1109/MSP.2017.2743240). URL: <https://ieeexplore.ieee.org/document/8103164/> (visited on 12/10/2025).
- [4] Koji Azuma, Sophia E. Economou, David Elkouss, Paul Hilaire, Liang Jiang, Hoi-Kwong Lo, and Ilan Tzitrin. “Quantum repeaters: From quantum networks to the quantum internet.” In: *Reviews of Modern Physics* 95.4 (Dec. 2023). ISSN: 1539-0756. DOI: [10.1103/revmodphys.95.045006](https://doi.org/10.1103/revmodphys.95.045006). URL: <http://dx.doi.org/10.1103/RevModPhys.95.045006>.
- [5] Charles H. Bennett, Gilles Brassard, Sandu Popescu, Benjamin Schumacher, John A. Smolin, and William K. Wootters. “Purification of Noisy Entanglement and Faithful Teleportation via Noisy Channels.” In: *Physical Review Letters* 76.5 (Jan. 29, 1996), pp. 722–725. ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.76.722](https://doi.org/10.1103/PhysRevLett.76.722). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.76.722> (visited on 01/14/2026).
- [6] Charles H. Bennett, David P. DiVincenzo, John A. Smolin, and William K. Wootters. “Mixed-state entanglement and quantum error correction.” In: *Physical Review A* 54.5 (Nov. 1996), 3824–3851. ISSN: 1094-1622. DOI: [10.1103/physreva.54.3824](https://doi.org/10.1103/physreva.54.3824). URL: <http://dx.doi.org/10.1103/PhysRevA.54.3824>.
- [7] Panagiotis Botsinis, Dimitrios Alanis, Zunaira Babar, Hung Viet Nguyen, Daryus Chandra, Soon Xin Ng, and Lajos Hanzo. “Quantum Search Algorithms for Wireless Communications.” In: *IEEE Communications Surveys & Tutorials* 21.2 (2019), pp. 1209–1242. ISSN: 1553-877X. DOI: [10.1109/COMST.2018.2882385](https://doi.org/10.1109/COMST.2018.2882385). URL: <https://ieeexplore.ieee.org/document/8540839/> (visited on 02/24/2026).
- [8] H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller. “Quantum Repeaters: The Role of Imperfect Local Operations in Quantum Communication.” In: *Phys. Rev. Lett.* 81 (26 1998), pp. 5932–5935. DOI: [10.1103/PhysRevLett.81.5932](https://doi.org/10.1103/PhysRevLett.81.5932). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.81.5932>.
- [9] Angela Sara Cacciapuoti, Marcello Caleffi, Rodney Van Meter, and Lajos Hanzo. “When Entanglement Meets Classical Communications: Quantum Teleportation for the Quantum Internet.” In: *IEEE Transactions on Communications* 68.6 (June 2020), pp. 3808–3833. ISSN: 1558-0857. DOI: [10.1109/TCOMM.2020.2978071](https://doi.org/10.1109/TCOMM.2020.2978071). URL: <https://ieeexplore.ieee.org/document/9023997> (visited on 09/02/2025).

- [10] Marcello Caleffi, Angela Sara Cacciapuoti, and Giuseppe Bianchi. “Quantum internet: from communication to distributed computing!” In: *Proceedings of the 5th ACM International Conference on Nanoscale Computing and Communication*. NANOCOM '18. ACM, Sept. 2018, 1–4. DOI: [10.1145/3233188.3233224](https://doi.org/10.1145/3233188.3233224). URL: <http://dx.doi.org/10.1145/3233188.3233224>.
- [11] Aparimit Chandra, Wenhan Dai, and Don Towsley. *Scheduling Quantum Teleportation with Noisy Memories*. May 12, 2022. DOI: [10.48550/arXiv.2205.06300](https://doi.org/10.48550/arXiv.2205.06300). arXiv: [2205.06300\[quant-ph\]](https://arxiv.org/abs/2205.06300). URL: <http://arxiv.org/abs/2205.06300> (visited on 01/14/2026).
- [12] Corning Incorporated. *SMF-28[®] ULL Optical Fibers | Ultra-low-loss Fiber*. Corning. URL: <https://www.corning.com/optical-communications/worldwide/en/home/products/fiber/optical-fiber-products/smf-28-ull.html> (visited on 02/23/2026).
- [13] Davies, Álvaro G. Bethany Iñesta, and Stephanie Wehner. “Entanglement buffering with two quantum memories.” In: *Quantum* 8 (Sept. 3, 2024). Publisher: Verein zur Förderung des Open Access Publizierens in den Quantenwissenschaften, p. 1458. DOI: [10.22331/q-2024-09-03-1458](https://doi.org/10.22331/q-2024-09-03-1458). URL: <https://quantum-journal.org/papers/q-2024-09-03-1458/> (visited on 09/02/2025).
- [14] Jeroen Dehaene, Maarten Van Den Nest, Bart De Moor, and Frank Verstraete. “Local permutations of products of Bell states and entanglement distillation.” In: *Physical Review A* 67.2 (Feb. 28, 2003), p. 022310. ISSN: 1050-2947, 1094-1622. DOI: [10.1103/PhysRevA.67.022310](https://doi.org/10.1103/PhysRevA.67.022310). URL: <https://link.aps.org/doi/10.1103/PhysRevA.67.022310> (visited on 09/17/2025).
- [15] Fu-Guo Deng, Gui Lu Long, and Xiao-Shu Liu. “Two-step quantum direct communication protocol using the Einstein-Podolsky-Rosen pair block.” In: *Physical Review A* 68.4 (Oct. 2003). ISSN: 1094-1622. DOI: [10.1103/physreva.68.042317](https://doi.org/10.1103/physreva.68.042317). URL: <http://dx.doi.org/10.1103/PhysRevA.68.042317>.
- [16] David Deutsch, Artur Ekert, Richard Jozsa, Chiara Macchiavello, Sandu Popescu, and Anna Sanpera. “Quantum Privacy Amplification and the Security of Quantum Cryptography over Noisy Channels.” In: *Physical Review Letters* 77.13 (Sept. 1996), 2818–2821. ISSN: 1079-7114. DOI: [10.1103/physrevlett.77.2818](https://doi.org/10.1103/physrevlett.77.2818). URL: <http://dx.doi.org/10.1103/PhysRevLett.77.2818>.
- [17] W. Dür, H.-J. Briegel, J. I. Cirac, and P. Zoller. “Quantum repeaters based on entanglement purification.” In: *Physical Review A* 59.1 (Jan. 1999), 169–181. ISSN: 1094-1622. DOI: [10.1103/physreva.59.169](https://doi.org/10.1103/physreva.59.169). URL: <http://dx.doi.org/10.1103/PhysRevA.59.169>.
- [18] W. Dür, M. Hein, J. I. Cirac, and H.-J. Briegel. “Standard forms of noisy quantum operations via depolarization.” In: *Physical Review A* 72.5 (Nov. 22, 2005), p. 052326. ISSN: 1050-2947, 1094-1622. DOI: [10.1103/PhysRevA.72.052326](https://doi.org/10.1103/PhysRevA.72.052326). URL: <https://link.aps.org/doi/10.1103/PhysRevA.72.052326> (visited on 09/29/2025).
- [19] Karim S. Elsayed, Wasiur R. KhudaBukhsh, and Amr Rizk. “On the Trade-off between Fidelity and Latency for the Quantum Link Layer with few Memories and Entanglement Purification.” In: *2024 International Conference on Quantum Communications, Networking, and Computing (QCNC)*. 2024 International Conference on Quantum Communications, Networking, and Computing (QCNC). July 2024, pp. 17–24. DOI: [10.1109/QCNC62729.2024.00013](https://doi.org/10.1109/QCNC62729.2024.00013). URL: <https://ieeexplore.ieee.org/document/10628208> (visited on 09/07/2025).

- [20] Michael Epping, Hermann Kampermann, Chiara macchiavello, and Dagmar Bruß. “Multi-partite entanglement can speed up quantum key distribution in networks.” In: *New Journal of Physics* 19.9 (Sept. 2017), p. 093012. ISSN: 1367-2630. DOI: [10.1088/1367-2630/aa8487](https://doi.org/10.1088/1367-2630/aa8487). URL: <http://dx.doi.org/10.1088/1367-2630/aa8487>.
- [21] Nicolas Heess et al. *Emergence of Locomotion Behaviours in Rich Environments*. 2017. arXiv: [1707.02286](https://arxiv.org/abs/1707.02286) [cs.AI]. URL: <https://arxiv.org/abs/1707.02286>.
- [22] IBM Quantum. *Quantum Teleportation*. IBM Quantum Learning, accessed March 2, 2026. International Business Machines Corporation. 2024. URL: <https://quantum.cloud.ibm.com/learning/en/modules/computer-science/quantum-teleportation> (visited on 03/02/2026).
- [23] Sarah Jansen, Kenneth Goodenough, Sébastien de Bone, Dion Gijswijt, and David Elkouss. “Enumerating all bilocal Clifford distillation protocols through symmetry reduction.” In: *Quantum* 6 (May 19, 2022), p. 715. ISSN: 2521-327X. DOI: [10.22331/q-2022-05-19-715](https://doi.org/10.22331/q-2022-05-19-715). arXiv: [2103.03669](https://arxiv.org/abs/2103.03669)[quant-ph]. URL: <http://arxiv.org/abs/2103.03669> (visited on 09/22/2025).
- [24] Karim S. Elsayed, Anam Tahir, Amr Rizk. *On Purification Strategies for Teleportation Fidelity for Quantum Communication Applications*.
- [25] Stefan Krastanov, Victor V. Albert, and Liang Jiang. “Optimized Entanglement Purification.” In: *Quantum* 3 (Feb. 2019), p. 123. ISSN: 2521-327X. DOI: [10.22331/q-2019-02-18-123](https://doi.org/10.22331/q-2019-02-18-123). URL: <http://dx.doi.org/10.22331/q-2019-02-18-123>.
- [26] Michael A. Nielsen and Isaac L. Chuang. *Quantum computation and quantum information*. 10th anniversary edition. Cambridge: Cambridge university press, 2010. ISBN: 978-1-107-00217-3.
- [27] Liangzhong Ruan, Brian T. Kirby, Michael Brodsky, and Moe Z. Win. “Efficient entanglement distillation for quantum channels with polarization mode dispersion.” In: *Physical Review A* 103.3 (Mar. 19, 2021), p. 032425. ISSN: 2469-9926, 2469-9934. DOI: [10.1103/PhysRevA.103.032425](https://doi.org/10.1103/PhysRevA.103.032425). URL: <https://link.aps.org/doi/10.1103/PhysRevA.103.032425> (visited on 09/22/2025).
- [28] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. *Proximal Policy Optimization Algorithms*. Aug. 28, 2017. DOI: [10.48550/arXiv.1707.06347](https://doi.org/10.48550/arXiv.1707.06347). arXiv: [1707.06347](https://arxiv.org/abs/1707.06347)[cs]. URL: <http://arxiv.org/abs/1707.06347> (visited on 09/02/2025).
- [29] *Stable-Baselines3 Docs - Reliable Reinforcement Learning Implementations; Stable Baselines3 2.8.0a4 documentation* — stable-baselines3.readthedocs.io. <https://stable-baselines3.readthedocs.io/en/master/>. [Accessed 13-03-2026].
- [30] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. Second. Cambridge, MA: The MIT Press, 2018. ISBN: 9780262039246. URL: <http://www.incompleteideas.net/book/the-book-2nd.html>.
- [31] François-Lavet Vincent, Henderson Peter, Islam Riashat, G. Bellemare Marc, and Pineau Joelle. “An Introduction to Deep Reinforcement Learning.” In: *Foundations and Trends® in Machine Learning* 11.3–4 (Dec. 2018), 219–354. ISSN: 1935-8245. DOI: [10.1561/22000000071](https://doi.org/10.1561/22000000071). URL: <http://dx.doi.org/10.1561/22000000071>.

- [32] Julius Wallnöfer, Alexey A. Melnikov, Wolfgang Dür, and Hans J. Briegel. “Machine Learning for Long-Distance Quantum Communication.” In: *PRX Quantum* 1 (1 2020), p. 010301. DOI: [10.1103/PRXQuantum.1.010301](https://doi.org/10.1103/PRXQuantum.1.010301). URL: <https://link.aps.org/doi/10.1103/PRXQuantum.1.010301>.
- [33] Christopher Watkins. “Learning From Delayed Rewards.” In: (Jan. 1989). URL: https://www.researchgate.net/publication/33784417_Learning_From_Delayed_Rewards.
- [34] W. K. Wootters and W. H. Zurek. “A single quantum cannot be cloned.” In: *Nature* 299.5886 (Oct. 1982). Publisher: Nature Publishing Group, pp. 802–803. ISSN: 1476-4687. DOI: [10.1038/299802a0](https://doi.org/10.1038/299802a0). URL: <https://www.nature.com/articles/299802a0> (visited on 02/23/2026).